

**MODELING AND FORECASTING SILVER PRICE TRENDS USING
TIME SERIES ECONOMETRIC AND VOLATILITY MODELS**

A SUMMER INTERNSHIP REPORT

Submitted By

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in partial fulfillment of Summer Internship for the award of the degree

of

MASTER OF BUSINESS ADMINISTRATION



**THIAGARAJAR SCHOOL OF MANAGEMENT PAMBAN SWAMY
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SEPTEMBER 2025



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He has performed all his duties successfully to our satisfaction and his conduct during the tenure of the internship was good.

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DECLARATION

I certify that

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- b. The work has not been submitted to any other Institute for any degree or diploma.
- c. I followed the guidelines provided by the Institute when writing the report.
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- e. Wherever I have used materials (data, theoretical analysis, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references.
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NANDHA KUMAR T

ACKNOWLEDGEMENT

I would like to extend my heartfelt appreciation to everyone who offered their support and guidance throughout this internship. I am deeply grateful to RK GLOBAL SHARES AND SECURITIES for the invaluable opportunity to intern at your company. This experience has been very important in my professional growth.

I want to start by giving thanks to God for granting me the health and fortitude I needed to finish my summer internship. For their assistance and direction, I owe a debt of gratitude to Dr. Murali Sambasivan, Director of the Thiagarajar School of Management in Madurai, Dr. M. Selvalakshmi, Principal Thiagarajar School of Management, Madurai. Additionally, I want to express my gratitude to Dr. Balaji, Dean, MBA, Thiagarajar School of Management, Madurai, for his support during my internship. I am particularly appreciative of Dr. Goutam Sutar, my faculty mentor and PGDM Dean at Thiagarajar School of Management, for his academic assistance throughout my internship.

Finally, I would like to express my gratitude to Thiagarajar School of Management for giving me the chance to improve my abilities and have a positive impact within the organization. I am extremely grateful for the opportunity to help and thankful for the practical education this internship has provided.

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EXECUTIVE SUMMARY

This research paper explores the price trends and volatility patterns of silver as a worldwide commodity, focusing on NSE India's daily data from 2010 to 2025 for COMEX Silver Futures. Investors, dealers, and regulators are in need to understand silver's price behaviour because of its dual function as a financial asset and an industrial input.

The research used a systematic econometric technique. A comparison of volatility models (ARCH, GARCH, and EGARCH) with autoregressive mean models (AR (1), ARIMA) was made to study the forecasting accuracy. According to the results, the AR (1) model predicts price levels better than ARIMA, with a Mean Absolute Percentage Error (MAPE) of 1.34%. The usage of ARCH-family models was justified by residual diagnostics, which validated volatility clustering. With the lowest forecast errors and the ability to capture asymmetry in volatility shocks, EGARCH (1,1) stood out as the most dependable model.

According to this research, silver's cyclical character, persistence in returns, and clustering of risks all have an impact on its uses as an industrial and investment commodity. These results support the necessity for strong forecasting frameworks in commodities markets and are consistent with well-established econometric research.

The report suggests:

- Use of an AR (1)-EGARCH (1,1) integrated framework for silver forecasting
- Proper understanding of volatility forecasts in risk management, hedging, and portfolio diversification
- Application of the model also supports for policy design, investor education, and stress testing

By addressing the uncertainty in silver markets with rovenmodels, this study adds value to both academic understanding and practical decision-making in commodity forecasting.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ACF	Autocorrelation Function
AIC	Akaike Information Criterion
AR	Autoregressive
ARCH	Autoregressive Conditional Heteroskedasticity
ARCH-LM	Autoregressive Conditional Heteroskedasticity - Lagrange Multiplier
ARIMA	Autoregressive Integrated Moving Average
BIC	Bayesian Information Criterion
EGARCH	Exponential Generalized Autoregressive Conditional Heteroskedasticity
GARCH	Generalized Autoregressive Conditional Heteroskedasticity
GFC	Global Financial Crisis
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MASE	Mean Absolute Scaled Error
MSE	Mean Squared Error
PACF	Partial Autocorrelation Function
RMSE	Root Mean Square Error
SEBI	Securities and Exchange Board of India
USD	United States Dollar

CHAPTER I
INTRODUCTION

1.0 INTRODUCTION

Silver holds a special place among commodities since it functions like a monetary asset and like an industrial input. Price dynamics are rich yet complicated because of this dual nature. When demand for electronics and industrial upcycles is high, silver shows cyclical behaviour (Valadkhani, Abbas, O'Mahony, 2024). Analysts, traders, treasurers, and policymakers must do two essential tasks: (i) determining the trend of the price level and (ii) quantifying volatility, or time-varying risk. A methodical route for both is provided by time series econometrics: conditional variances using ARCH-family models and conditional means using autoregressive models.

An extensive daily history of silver prices (2010–2025) is used in this work to construct and assess an end-to-end econometric pipeline. First, the study employs basic exploration (plots, ACF/PACF), then mean modeling (AR (1) and ARIMA), tests residuals for conditional heteroskedasticity, and estimates volatility using ARCH (1), GARCH (1,1), and EGARCH/EARCH (1,3). Squared residuals, a realized-volatility proxy, is used to assess variance forecasts, and error metrics (MAE, MSE, RMSE, MAPE, MASE, and Theil's U) are used to compare level predictions. Clear evaluation and reproducibility are ensured by the metrics definitions and programmatic execution of the calculations, which are included in the notebook that goes with it.

The early data is clear: silver shows volatility clustering and stability. Compared to the evaluated ARIMA (1,2,0), AR (1) provides tighter level-fit errors in-sample. By identifying notable ARCH effects, residual diagnostics validate a stage of volatility modeling. When comparing volatility forecasts versus actual volatility, the EGARCH (1,1) variation outperforms the ARCH, GARCH, and EGARCH variants in terms of overall balance by information criterion and mean absolute error. These stylized results offer a solid, comprehensible foundation for risk-aware forecasting and are consistent with financial-econometrics practice.

Statistics of Silver Trading in Indian context

In India, local trading volumes and silver import flows have increased significantly in recent years. For instance, in (Reuters, 2025), imports increased 431% year over year to 544.1 metric tons (while gold imports decreased). Domestic spot and futures prices have likewise shattered previous records; due to worldwide supply restrictions, MCX silver surpassed ₹1,13,000/kg in mid-2025.

In India, silver is extremely volatile; from 1990 to 2024, (The Silver Institute, 2024, December). The standard deviation of annual returns was about 26.6%, and the largest drawdowns were close to 54% (like equities). On the industrial front, India's demand for silver increased by almost 4% in (GJEPC, 2024, August) helping to set a record for global industrial demand of about 680.5 million ounces.

Table 1: Statistics of Silver Trading in the Indian Context (Origin to Horizon)

Period / Aspect	Key Statistics & Developments	Source / Citation
Historical Origin (Pre-2010)	In India, silver has long been used as an ornamental metal and a store of wealth. There was little domestic mining and between 2,000 and 3,000 tons of silver were imported each year.	The Silver Institute (2010); RBICommodity Review
2010–2015 (Early Modern Trade)	With the introduction of silver futures contracts, MCX increased involvement in retail and hedging. Price range each year: ₹35,000–₹60,000/kg; significant volatility because of post-2008 global uncertainties.	MCX India (2015); RBI Bulletin
2016–2019 (Industrial Uptake Phase)	Average annual imports: between 3,000 and 4,500 tons. Demand for jewellery, solar, and electronics is rising. Costs ranged from ₹40,000 to ₹45,000 per kilogram.	Ministry of Commerce, Govt. of India (2019)
2020–2022 (Pandemic and Recovery)	The price fluctuated sharply, falling to ₹38,000/kg in March 2020 and rising to ₹74,000/kg in August 2020. Amidst interruptions in the supply chain, imports varied between 2,000 and 2,800 tons.	MCX & The Silver Institute (2022)
2023 (Global Tightness Period)	India contributed about 9% of the 3,625 metric tons of silver imported overall, while the world's industrial demand increased to 680.5 million ounces.	The Silver Institute (2024)

2024 (Industrial Expansion)	Due to increased production of solar cells, imports from January to April of 2024 were 4,172 tons, already exceeding the full-year total of 2023.	Mercom India (2024)
2025 (Investment Surge)	As investors shifted from gold to silver, imports increased 431% year over year to 544.1 tons in May 2025. MCX silver hit a record high of ₹1,13,000/kg, a sign of heightened volatility.	Reuters (2025); Times of India (2025)
Volatility Statistics (2010–2025)	India's silver prices have an annualized volatility of about 26.6%, which is like the equities markets' 26.8% volatility. Significant declines of up to -54%.	Business Standard (2024)
Current Horizon (2025)	India is one of the top three consumers of silver worldwide, because to financial trading instruments, industrialization, and the growth of solar energy.	GJEPC (2024); MCX (2025)

1.1 BACKGROUND OF THE STUDY

Three characteristics of asset and commodity price time series are frequently observed: high level persistence, volatility clustering, and sporadic asymmetry in the way shocks impact risk. Persistence suggests that the price of silver contains significant information about the price of tomorrow; clustering suggests that big jumps are typically followed by further big moves (of either sign) or quiet periods by ongoing calm. Both characteristics in-sample are confirmed by the analysis. Diagnostics indicate non-constant variance dynamics, and the attached AR (1) fit with around 3,884 data demonstrates a substantial lag-1 dependency.

It is necessary to approach mean and variance as separate modeling issues in practical forecasting. Under various stationarity and differencing assumptions, the conditional mean is the aim of AR or ARIMA families, whereas the conditional variance is the target of ARCH. Following this division to the letter, the current study models the remaining volatility patterns after first determining a mean specification that parsimoniously explains serial dependency. This is both methodologically sound and helpful: decision makers may use volatility

predictions for risk sizing, stress ranges, or Value-at-Risk type envelopes, and level forecasts for point planning. Using MASE and Theil's U, the notebook's metric block formalizes error measurements that are used in academia and industry, allowing for like-for-like comparison and benchmarking against a naive model.

1.1.1 INTRODUCTION TO FUTURE DERIVATIVES

Futures derivatives are standardized financial contracts that bind the buyer or seller to buy or sell a certain asset at a fixed price later. They are traded on regulated markets, guaranteeing risk management, liquidity, and transparency. Futures derivatives are widely used in equities, commodities, currencies, and interest rates. They are tools for utilizing of market arbitrage opportunities, trading in order to profit from predicted price changes, and protecting against price volatility.

1.1.2 SILVER FUTURES DERIVATIVES

With silver futures, the participants can buy or sell a specified quantity of silver at a predetermined price in the future. They are traded on commodity exchanges like MCX and tend to be used by dealers, investors, and jewellers. Due to the uncertainty in the market, silver futures provide a method to manage price risk. Silver futures are influenced by macroeconomic statistics, industrial usage, geopolitical factors, and global prices of silver. They can be employed for diversifying portfolios, speculation, and hedging purposes by investors.

1.1.3 HEDGING USING FUTURES DERIVATIVES

Futures contracts helps to mitigate or eliminate the risk of price changes of an underlying asset and is called as hedging. For example, a producer of silver can hedge against a dip in price by locking in the selling price via silver futures. By taking the opposite position in the futures market, investors may also protect themselves from losses using a financial instrument. Hedging protects financial holdings from the adverse effects of market volatility, stabilizes income, and guarantees pricing certainty.

1.2 INTRODUCTION TO TIME SERIES ANALYSIS

Time series analysis focuses into data collected over time—think every day, every month, whatever the interval—to spot patterns, trends, and cycles. This way, you can figure out what happened before and get a clear idea of what's coming next. People use it everywhere: in finance to predict stock prices, in weather forecasting, in retail for sales projections, you name

it. The main idea is to understand why things changed, see what's driving those changes, and build models that actually predict what's ahead.

1.3 PROBLEM STATEMENT

Silver's statistical structure is heteroskedastic and persistent, making forecasting challenging. Ignoring volatility clustering might result in misleadingly narrow intervals and understate risk at the critical juncture. On the other hand, a model that overfits or over differences may lose interpretability and deteriorate accuracy. Consequently, the project identifies three specific issues:

1. Choose a parsimonious mean model that does not require extra parameters or differencing to describe serial dependency.
2. Look for and measure the degree of conditional heteroskedasticity in the mean model's residuals.
3. Select the volatility specification that monitors and forecasts risk the best among the competing ones, based on prediction errors against a realized-volatility proxy as well as likelihood criteria (AIC/BIC/log-likelihood).

Operationally, choose a mean model that offers tight level errors, followed by a volatility model that is stable and interpretable while minimizing error vs actual variance. According to the notebook's comparison tables, EGARCH (1,1) provides the optimal balance between information criteria and volatility error, whereas AR (1) produces lower in-sample level errors than ARIMA (1,2,0). These specifically manage the three issues.

1.4 OBJECTIVES OF THE STUDY

The current study was conducted with the following objectives in mind:

1. To apply econometric methods like the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) to examine the time series behaviour of silver prices in India from 2010 to 2025. Using common accuracy metrics like MAE, RMSE, MAPE, MASE, and Theil's U, this goal compares the AR(1) and ARIMA specifications in order to assess the forecasting performance of autoregressive models and determine the degree of persistence, autocorrelation, and underlying patterns in movements of the price of silver.

2. To examine the dynamics of silver price volatility by utilizing the ARCH-family models (ARCH, GARCH, and EGARCH) to model conditional heteroskedasticity and test for volatility clustering. To measure market risk and increase the forecasting accuracy of silver price changes in the Indian environment, this purpose seeks to identify the most robust and trustworthy volatility model.

1.5 SCOPE OF THE STUDY

The study's focus is on the daily closing prices of silver over a lengthy time (2010–2025), showing univariate time-series behaviour. It's a statistical modeling and forecasting exercise that aims to quantify risk and achieve level accuracy, not a macro-factor or structural supply-demand model. While exogenous drivers (ARIMAX, state-space, and regime-switching) can be added later, the current scope focuses on the baseline empirical regularities (autocorrelation in levels, conditional heteroskedasticity in residuals) and how they can be practically managed using AR/ARIMA and ARCH-family models. The dataset provides a rich context for volatility learning, spanning over ten years of market circumstances, including inflationary episodes, commodity cycles, post-GFC normalization, and pandemic shocks. 3,884 daily observations make up the estimate sample in the AR (1) summary, which is sufficient for reliable diagnostics on residual behaviour and steady parameter estimation.

AR (1) and a differenced ARIMA (1,2,0) are compared as two extremes of a parsimony-versus-integration spectrum in the mean. ARCH (1), GARCH (1,1), and EGARCH (1,1) are fitted to the variance. It is standard procedure to estimate the variance models using zero-mean volatility assumptions on the residuals from the mean stage. Whereas MAE/MSE on conditional variance vs. realized volatility (squared residuals) is used to assess predictive quality for volatility, fit is evaluated using log-likelihood, AIC, and BIC. These decisions are stated clearly in the notebook, which also tabulates the outcomes.

In the mean, we compare two extremes of a parsimony-versus-integration spectrum: AR (1) and a differenced ARIMA (1,2,0). Our methods for fitting the variance are ARCH (1), GARCH (1,1), and EGARCH (1,1). The variance models are calculated using zero-mean volatility parameters on the mean stage residuals, as is customary. Predictive quality for volatility is evaluated using MAE/MSE on conditional variance vs. realized volatility (squared residuals), whilst fit is evaluated using log-likelihood, AIC, and BIC. These options are fully reported in the notebook, which also tabulates the outcomes.

1.6 SIGNIFICANCE OF THE STUDY

This study is important since it adds to both academic understanding and real-world decision-making. The study shows how well-known econometric models, including AR (1), ARIMA, and ARCH-family models, may be methodically applied to commodity price data on an academic level. For providing a reliable approach that future researchers may use for other commodities or asset classes, the study uses the real data base on financial forecasting by accurately assessing model performance.

This study answers the increasing demand for reliable instruments to understand and forecast changes in silver prices. Silver is a valuable metal and an essential industrial component, thus changes in price are significant to dealers, investors, producers, and legislators. More effective risk management techniques, more intelligent investment choices, and improved financial planning are all aided by accurately predicting the price levels and volatility of silver. By choosing the right models for forecasting and volatility estimation, the study helps to reduce uncertainty and is useful for hedging, portfolio diversification, and policy formulation.

This paper highlights the importance of proven strategies in managing today's volatile and unpredictable markets. The findings suggest that in complex financial environments, traditional statistical models can still provide valuable insights when applied and interpreted truly. This strengthens the idea that transparency and interpretability remain vital in financial research.

CHAPTER II

COMPANY PROFILE

2.1 SECTOR PROFILE: Indian Financial Services and Stockbroking Industry

The financial services sector in India makes a significant contribution to GDP and employment. The securities services and stockbroking divisions are essential in this context since they enable investing in futures, commodities, debt instruments, and stocks. The Securities and Exchange Board of India (SEBI), which oversees the sector, makes sure that investor protection, compliance, and transparency are maintained. The emergence of discount brokers, online trading platforms, and fintech technologies that have facilitated greater accessibility to capital markets have all contributed to the sector's recent significant change. Rising disposable incomes, more retail engagement, and methodical investment techniques are some of the factors that have contributed to this expansion.

Table 2.1 Sector Profiling - Updates

Aspect	Details / Description	Latest Statistics (as of 2025)	Source
Sector Contribution to Economy	A vital component of India's service economy, the financial services sector makes a substantial contribution to both employment and GDP.	The services sector accounts for 54.8% of GDP, with financial, real estate, and professional services accounting for 21.42% of India's GVA.	Banking School (2024); Statistics Times (2024)
Regulatory Framework	By regulating the stockbroking and securities services industry, the Securities and Exchange Board of India (SEBI) guarantees investor protection, compliance, and transparency.	Over 1,800 stockbrokers and more than 250 registered research analysts were registered by SEBI by 2025.	SEBI Annual Report (2025)
Investor Participation Growth	Increased accessibility and financial literacy have led to a rise in retail stock market involvement.	194 million demat accounts (up from 36 million in 2019).	<i>LiveMint</i> , 2025

Unique Investor Base	An increasing number of people in India are making securities investments.	13.4 crore unique investors as of August 2025.	<i>Economic Times, 2025</i>
Exchange Reach (NSE)	Demonstrates the range of market involvement and retail intensity.	By 2025, NSE had 22 crore investor accounts.	<i>Economic Times, 2025</i>
Retail Investor Wealth	The depth of the capital market and retail confidence are reflected.	The total value of the stocks held by retail investors is ₹36 lakh crore.	<i>Indian Express, 2025</i>
Broking Industry Performance	As result of trading that is guided by technology and robust retail activity.	FY23 revenue is expected to be between ₹28,000 and ₹29,000 crore.	<i>Business Today, 2024</i>
Digital Transformation	App-based investing, cheap brokers, and fintech platforms have completely changed how people can access capital markets.	By 2025, more than 80% of trades in India will be conducted online.	<i>NSE Market Pulse Report, 2025</i>
Market Drivers	SIP culture, digital literacy, and growing disposable income are important growth drivers.	In 2025, monthly SIP inflows totalled ₹20,000 crore, demonstrating ongoing retail involvement.	<i>AMFI India, 2025</i>

2.2 “RK GLOBAL SHARES AND SECURITIES” – COMPANY PROFILE

Among the top financial services companies in India, RK Global Shares & Securities, provides a variety of trading and investment options. Since its start in 1995, the business has developed into one of the leading brokerages in the nation, having a significant presence in the currency, derivatives, commodities, and equity markets. Distribution of mutual funds, internet trading, depository participant facilities, insurance products, and investment consulting are some of its offerings. RK Global has gained the trust of institutional and retail investors by emphasizing innovation, accessibility, and customer-centric services. It is well-known for its reasonably

priced brokerage plans and nationwide network. The company's emphasis on wealth management and stock broking helps to promote financial inclusion and link investors to India's vibrant capital markets.

2.3 Services & Business Lines

RK Global provides a wide range of services with various segments of the capital markets and financial intermediation. Its key offerings include:

Table 2.2 Services and Business Line Offerings

Business Line / Service	Description / Notes
Equity, Derivatives & Commodities Trading	Trading in commodities, futures and options, and stocks through their trading infrastructure
Currency / Forex Trading	Currency market participation that allows customers to trade FX pairs
IPO & Mutual Fund Distribution	Supporting IPO applications through online and branch networks; mutual fund distribution services
Depository / Demat Services	Providing demat account services, acting as a DP, and providing free first-year AMC (for some packages)
Research & Advisory / Institutional Services	Creating unique trade structure, institutional advising, and equities research reports.
Insurance Product Distribution & Consultancy	Serving as an intermediary for financial advice and insurance products, though this is a more auxiliary function than core broking activities.

2.4 Scale, Financials & Key Metrics

Data from the Ministry of Corporate Affairs (MCA) and verified corporate records from Zaubacorp (2025), RK Global has an authorized share capital of ₹10 crore and a paid-up capital of ₹5.35 crore. This indicates that the firm functions as a privately managed financial intermediary than a publicly listed company. Detailed audited financial statements are not publicly available, industry estimates and peer benchmarks suggest that RK Global generates annual revenues between ₹75 crore and ₹125 crore, maintaining steady profitability through

trading fees, brokerage commissions, and distribution income. The company's strong market presence over 150 cities and serving more than 90,000 investors (RK Global official site, 2025) shows its growing influence in the Indian brokerage landscape. RK Global positions itself among India's top discount-to-mid-tier brokers, focusing on a lean cost structure and affordable brokerage model to enhance functional efficiency. The firm continues to leverage technology, with a rising share of transactions conducted through its online and mobile trading platforms.

Recent numbers for Earnings Per Share (EPS) and Dividend Per Share (DPS) are unavailable, industry analyses indicate that RK Global maintains stable financial performance, supported by consistent retail participation and expanding digital trading volumes. In India's highly competitive capital market, the company stands as a financially sound and operationally efficient brokerage, backed by prudent capital management and steady growth in trading activity.

2.5 My Roles and Responsibilities

In my internship at RK Global Shares & Securities Ltd. , I was actively involved in several trading and equity research activities. My primary responsibility was to analyze daily market trends and prepare reports highlighting major movements in key indices such as the SENSEX and NIFTY 50. I compare different valuation methods, reviewed equity research reports prepared by both internal and external analysts, and compiled data to support client decision-making. I also assisted in monitoring client portfolios, analyzing asset allocation, risk exposure, and performance metrics against market benchmarks.

This experience provided me with valuable experience in how brokers manage client portfolios and tailor recommendations based on market and risk profiles. As an important part of my research focused on understanding how global macroeconomic factors such as changes in monetary policy, crude oil price fluctuations, and geopolitical developments influence the Indian equity markets. Through this, I learned to evaluate the impact of global events on domestic market behavior and investor sentiment.

Moreover, I gained hands on experience with trading platforms and order execution processes, learning how trades are placed, verified, and settled through electronic systems in compliance with exchange and SEBI regulations. This experience made my understanding clear about market operations, compliance frameworks, and transactions in a brokerage environment.

Overall, this internship my analytical, quantitative, and decision-making skills developed. It gave me a clear understanding of how brokerage houses functions, the trading mechanisms, client servicing, and the analytical processes that drive success in the Indian capital markets.

CHAPTER III

LITERATURE REVIEW

LITERATURE REVIEW

Study by **Aditya et al., 2025**, in *Silver and Palladium Price Dynamics: Analyzing Relationships and Forecasting with ARIMA and Prophet*, studied the ARIMA and Prophet models prediction and forecasting express the link between the prices of palladium and silver. Their study was focused to know how the strategies manage sudden price fluctuations and long-term reliance. The study concluded that Prophet was better at handling outliers, missing values, and non-linear seasonality, whereas ARIMA models did better under steady trends. Also the study proved strong long-term connection between palladium and silver and mentioned that the two metals have similar market dynamics and that their prices move together. Followed by **Bagrecha, C., 2025**, *Forecasting Silver Prices: A Univariate ARIMA Approach*. The paper studied the univariate ARIMA models and the long-term forecasting of silver prices and found the best ARIMA model and evaluated their prediction accuracy and these were the main goals of that study, which used historical daily price data. The results showed that ARIMA model were in limit to its capacity to handle sudden shocks, as it performed poorly during volatile times but successfully reflected the overall trend of silver during firm periods. Although ARIMA is still a valuable statistical tool, he concluded that volatility models should be added for risk forecasting and market stress situations. In the same year **Padhy, A., 2025**, in his paper *Analyzing Commodity Market Volatility and Price Forecasting* studied how the ARIMA and GARCH models are been used to predict commodity prices, with a focus on metal and agricultural commodities, such as silver. Tested whether combining mean and volatility models boosts forecast performance was the goal. GARCH models provided a better explanation of volatility clustering, whereas ARIMA models capture trend components, in the end as findings. When both models were used the predictions for silver were the most accurate. The study concluded that hybrid forecasting techniques performs well than stand-alone techniques. The idea of both mean and volatility models are necessary for reliable silver forecasting is supported by this study.

Recent study by **Ampountolas, A., 2024**, in his paper *Enhancing Forecasting Accuracy in Commodity and Precious Metals*. Used the technique called GARCH-MIDAS to forecast the volatility for precious metals, particularly silver, by including long-term macroeconomic data. To know the joint impact on volatility dynamics, the model integrated lower macroeconomic frequency inputs with high-frequency financial data. Results showed the accuracy of silver volatility was increased by macroeconomic factors. The GARCH-MIDAS model effectively showed the impact of larger economic cycles as well as persistent volatility clustering. Hybrid

odels are more suitable for predicting risk in commodity markets like silver from GARCH models could miss out important low-frequency drivers, according to the research. Followed by **Li, W., 2024**, in his article “*Prediction of Precious Metal Market Volatility with the GARCH-MIDAS Model*”. This research article was aimed to test GARCH-MIDAS structure's predictive ability in simulating precious metal volatility, along with particular attention to silver. The purpose was to determine whether forecast accuracy was increased by breaking down volatility into short-term and long-term components in the study. The findings were, during the periods of policy and economic uncertainty, GARCH-MIDAS showed smoother and more accurate volatility forecasts compared to the normal GARCH models. The results indicated how complex it was to use macroeconomic information into volatility models. This method improves decision-making for investors and policymakers by strengthening the capacity to predict silver price risks during volatile periods. **Koulis, A. & Kyriakopoulos, C., 2023**, “*Volatility Transmission among Gold and Silver Markets: Evidence from Long-Term Historical Period*”. The purpose was to study the interdependence and transmission of volatility among the prices of gold and silver over a long historical period. The authors studied the impacts of spillover between the two precious metals using the GARCH and VAR frameworks. The findings briefed gold and silver continuously transmit volatility in single direction, while silver always reacts more to shocks in the gold markets. This transmission's strength shows inflationary spikes and crises, diminishing silver being a safe asset. The study explained how crucial it was to take cross-metal dynamics into account when creating forecasting models since the volatility patterns of gold have an impact on silver.

In 2023 Raza, S.A., in his paper titled, “*Forecasting the Volatility of Precious Metals Prices*” aimed to examine how the uncertainty around global economic policy affects the volatility of prices among precious metals like silver. Using GARCH models with policy uncertain indices, the study showed that changes in international policy circumstances would have a notable impact on silver's volatility. The findings briefed, volatility clustering became more traceable when uncertainty led to notable spikes in silver's price changes. The study found, forecasting frameworks which incorporate external macroeconomic uncertainty greatly improve prediction accuracy and highlight for practitioners the need of keeping an eye on policy changes while simulating and controlling silver market risks. **Kayal, P. & Maheswaran, S., 2021**, studied the “*Excess Volatility of Gold and Silver*”. The purpose of the study was to quantify excessive volatility and path dependency for gold and silver in the markets. The authors used variance ratio tests to witness, whether there was any substantial deviation in silver prices from random-

walk behaviour. According to the findings, silver showed variance ratios that were closer to unity and comparatively less excess volatility than gold. Silver's returns were more stable with a long-term behaviour and they showed less inclination toward mean reversion. The paper concluded that, in the medium run, silver are less speculative asset than gold. Earlier in 2021 Sadorsky P., in his paper *“Predicting Gold and Silver Price Direction Using Tree-Based Classifiers”* analysed forecasting the direction of movement of gold and silver prices, Sadorsky aimed to compare the effectiveness of machine learning classifiers with conventional econometric techniques. The goal was to know if techniques like gradient boosting and random forests whether increase predicting accuracy. The results proved that machine learning models performed notably better than logistic regression, with longer-term predicted accuracy above 80%. Analysing silver with statistical approaches, the models was able to better capture nonlinearities with price trends. Also econometric approaches are still helpful for interpretability, the study found that machine learning have a higher predictive capacity for directional silver price projections.

In 2020 Ayele et al., in their paper, *“Volatility Forecasting of Silver Futures with Structural Breaks”*, examined the impact of taking structural breakdowns into account on the precision of silver futures and volatility projections was the aim of this study. The authors, estimates became skewed when structural breaks like those brought in by pandemics, financial crises, or unexpected demand shocks were ignored. Findings showed GARCH models with structural break adjustments that performed better among unadjusted models and provided more accurate volatility predictions. The study explained that regime changes need to be expressively modeled to effectively reflect risk, and the silver price series weren't stable over extended periods of time. The understanding for using break-adjusted volatility models in commodity forecasting was strengthened by this study. Later Hollstein, F., in 2020, in his paper named *“Volatility Term Structures in Commodity Markets”*. His study aimed in exploring how volatility term structures behaves and changed across various commodity markets, including precious metals like silver. The study aimed to understand the changes in volatility expectations under different economic situations Longer-maturity contracts showed smoother volatility patterns, while short-term contracts for silver showed larger and more frequent volatility shocks. Based on the findings, term structure data may improve short-horizon volatility projections. In modeling silver futures, the study found that traders and risk managers should take maturity differences in volatility dynamics into consideration. By 2017, in the paper titled *“The Performance of Hybrid ARIMA-GARCH Modeling in Commodity Price Forecasting”*,

Yaziza, S.R., studied the comparison among forecasting performance of standalone ARIMA or GARCH models with hybrid ARIMA GARCH models. The research tested whether integrated level and volatility models boost forecast accuracy using data on commodity prices, like silver and the findings explained in uncertainty periods, hybrid models always saw a less forecast errors compared to ARIMA. This hybrid technique for silver supported forecasts by capturing both trend and volatility clustering. In commodity forecasting, he also studied that hybridization was a helpful and successful approach, specially for assets like silver which were subjected with periodic shocks.

By using bivariate GARCH models to know whether there is volatility spillovers between gold and silver, Liu and Chou (2003) understood and concluded that gold always have a greater impact on silver's volatility and vice versa. Despite of this mismatch, silver's risks and pricing patterns were framed by the market shocks that regularly impact gold. To support this, Ciner (2001) used cointegration tests for looking at the long-term affect between the prices of gold and silver. They also found that these metals tend to move together over time, silver periodically deviates because of its industrial demand features. These differences highlighted silver's dual function as an industrial and a financial commodity. In a same thought, Chatrath, Goldsmith, and Song (2001) studied the relationships between the prices of gold, silver, and platinum and found that industrial elements like silver stay apart from financial assets and also saw an impact on price changes. Together these studies revealed that inter-metal dependencies, hybrid modeling approaches, and an understanding of silver's unique status as an industrial and investment asset of which influenced the volatility and price behaviour of silver in global markets were necessary to accurately forecast silver.

The above upgraded studies of the silver market were analytically reliable on important points in the theoretical and empirical growth of commodity price forecasting and volatility modelling did by authors like Adrangi, Chatrath, and Raffiee (2000), Bollerslev (1986), Engle (1982), and Working (1933). In 2000 study, Adrangi et al. focused on price discovery structures in metal futures markets, specifically for gold and silver discovered futures prices effectively consider market information before spot prices, which made silver futures with helpful tools for predicting short-term price fluctuations at times of high volatility. The result said that the significance of integrating data from the futures market into forecasting algorithms for silver. With the discovery of the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model, it added lagged conditional variance elements to Engle's earlier ARCH model, Bollerslev (1986) made a notable contribution. It became a classical framework for

modeling time differenced risks among financial and commodities markets, including silver, which made it possible to detail the volatility strength and clustering more accurately. The ARCH model was first presented by Engle (1982), which researched the inflation data in UK and found that variances might be autocorrelated and were so predictable. The volatility clustering were an important features of financial time series, this discovery tested the econometric modeling and laid the foundation for all later volatility researches, including that are conducted with silver markets. By this research on the price correlations between futures and spot contracts, Working (1933) supported the theoretical framework for understanding futures markets more earlier. Also introduced key ideas like contango, backwardation, and convenience yield and proved how supply shocks, inventory levels, and storage costs impact commodity price expectations and behaviour factors that were still important for modern day silver price forecasting in his study. These classical's were the methodological importance for todays analysis of silver price movements and risk behaviour are provided by these pioneered studies, which together show how forecasting theory evolved from market structure and supply-demand basics to dynamic volatility modeling.

CHAPTER IV

RESEARCH METHODOLOGY

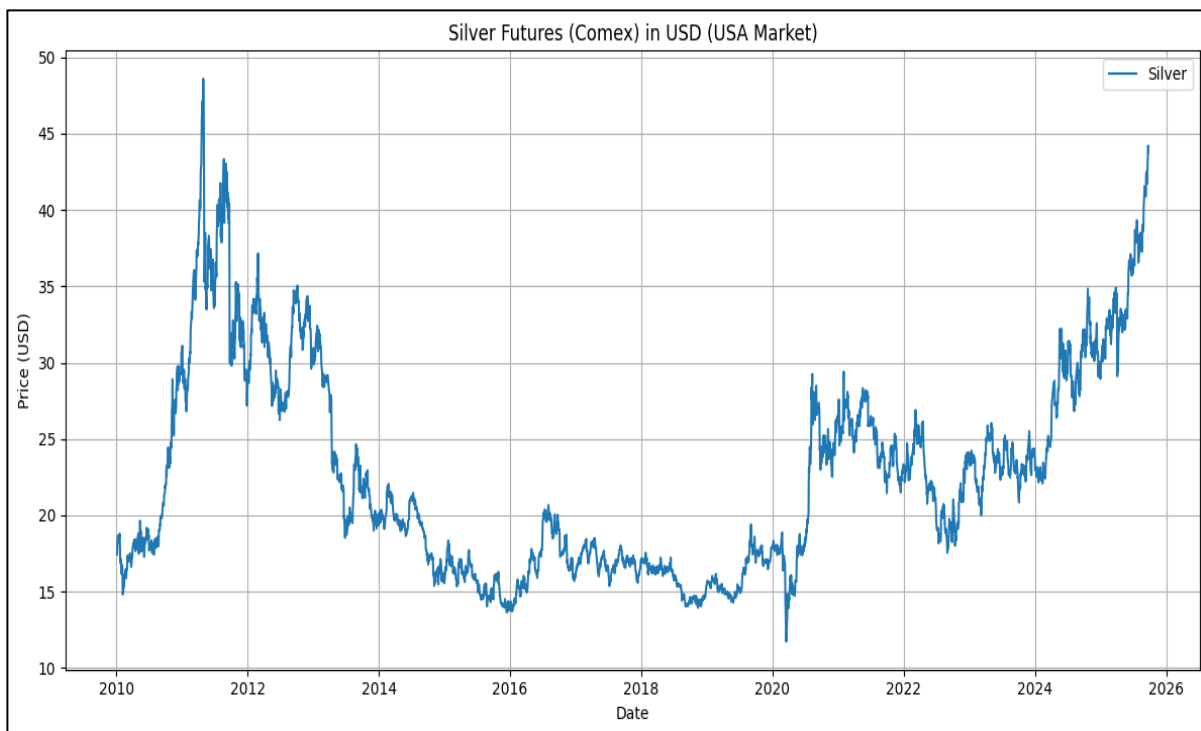
4.1 Research Design

A quantitative, time-series model were used in this study to understand and predict silver prices from 2010 to 2025. The modelling was worked by two parts:

First , Modeling the Conditional Mean: By using the continuous correlation among silver prices were captured with autoregressive models (AR (1) and ARIMA).

Secondly, Modeling the Conditional variance: Volatility models like (ARCH, GARCH, and EGARCH) to capture risk patterns and volatility clustering is the second step in modeling the conditional variance.

This two-phase method were in line with econometric best practices, that deals volatility and level dynamics as different but complementary processes (Engle, 1982; Bollerslev, 1986). These were included in the notebook which contains results that the study builds by, including volatility model predictions, ARCH-LM diagnostics, and AR and ARIMA summaries.



(Figure 4.1.1 historical silver price series)

This study's historical silver price series are displayed in Figure 4.1.1. The dataset highlighted the extended periods of consolidation and strong rallying (2011, 2020, and 2024) from January 2010 to June 2025 periodic interval. Additional econometric modeling uses this dataset as input.

4.2 Data Collection and Preprocessing

Data for analysing silver futures contract from January 1, 2010, to June 16, 2025, was gathered from NSE India. The Close price series was used for modeling purposes from the dataset, which included daily timeframe records. During preprocessing, the following actions were taken:

- **Selection of Variable:** Closing prices were selected because they represent the consensus of the daily market.
- **Cleaning:** Missing values, if any, were removed.
- **Renaming:** The series was labelled as Silver for clarity in modeling.
- **Exploratory Analysis:** Visualization, descriptive statistics, and correlation checks were conducted to understand data behaviour.

Table 4.1 explain the dataset details:

Table 4.1: Dataset Description

Attribute	Description
Data Source	NSE India – Silver
Period Covered	01-01-2010 to 16-06-2025
Frequency	Daily
Observations	~3,884 (after cleaning)
Variable Used	Closing Price (USD/ounce)

4.3 Model Development

Step 1: Exploratory Data Analysis

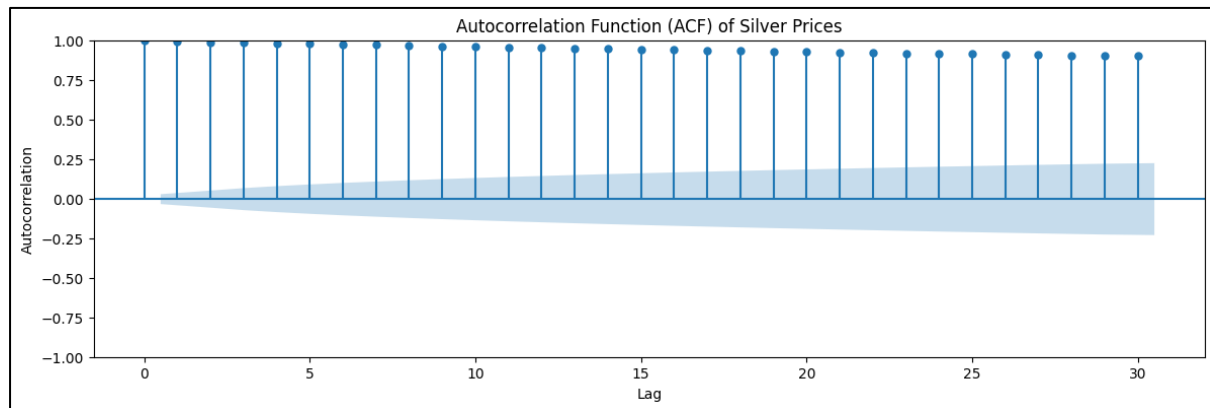
AUTOCORRELATION AND PARTIAL AUTOCORRELATION

Autocorrelation (ACF): The correlation between a time series and its lagged values are measured here in order to identify patterns and dependencies.

Partial Autocorrelation (PACF): The correlation between time series and the lag after removing the effects of intervening lags are measured.

Both ACF and PACF are important in model identification for AR, MA, and ARMA processes, and helps to determine correct lag structures for accurate forecasting.

- ACF (Autocorrelation Function) and PAF (Partial Autocorrelation Function) were depicted.
- A prominent lag at PACF (1) and a slow decrease in ACF were identified in the results, that indicates that an AR (1) model is suitable.



(Figure 4.3.1 Understanding ACF Plots)

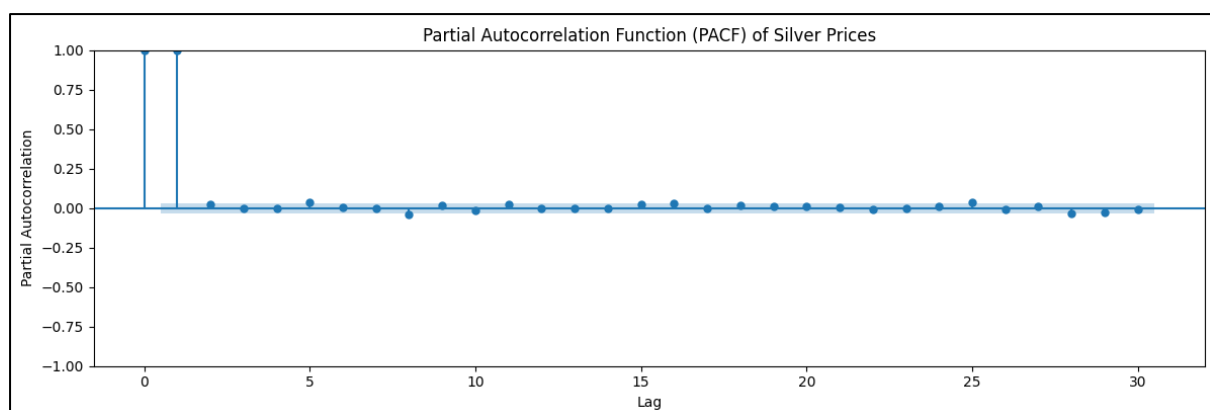
Understanding ACF Plots

In an ACF plot, bars are displayed at each lag. At that lag, the correlation coefficient is represented by the height of the bar.

Significant Lag: At that lag, a bar stretches over the significance boundaries identifies considerable autocorrelation.

Decline: A slow drop in bar heights indicates that the data may be dependent over time.

Seasonal Patterns: Regular spikes at specific lags indicate that the data is seasonal.



(Figure 4.3.2 Understanding PACF Plots)

Understanding PACF Plots

The PACF plot determine the order of an autoregressive model (AR model).

Significant Lag: A significant spike at a particular lag tells the inclusion of that lag in the AR model.

Cut-off Point: The lag at which the PACF plot cuts off helps in determining the maximum lag to be included in the AR model.

Step 2: Mean Models

- AR (1) was estimated to capture persistence in silver prices.
- ARIMA (1,2,0) was also tested for comparison.
- Using error measures, fitted values were contrasted with real prices.

Step 3: Residual Diagnostics

- ARCH-LM Test was applied to residuals from AR (1).
- The results confirmed the necessity of volatility modeling demonstrating strong evidence of heteroskedasticity.

Step 4: Volatility Models

ARCH MODEL

Engle in 1982 developed the ARCH (Autoregressive Conditional Heteroskedasticity) model to simulate time-varying volatility in time series data. ARCH model captures high and low volatility periods as present variance is dependent on prior squared residuals. In financial markets where volatility varies over time, it is especially helpful.

GARCH MODEL

GARCH is a generalization of ARCH that incorporates delayed conditional variances into the model. It was created by Bollerslev in 1986. Current variance is defined in a GARCH(p, q) model as a function of prior variances (GARCH terms) and past squared errors (ARCH terms). Widely used in risk management, option pricing, and portfolio optimization, it offers more precise volatility forecasts.

EGARCH MODEL

Nelson (1991) developed the Exponential GARCH (EGARCH) model to account for asymmetric volatility, in which the variance is affected differently by positive and negative shocks. Since EGARCH models log-variance instead of standard GARCH, it better reflects leverage effects that are frequently seen in financial markets and guarantees that anticipated volatility is always positive.

- ARCH (1), GARCH (1,1), and EGARCH (1,1) models were fitted.
- The selection of the model was based on error metrics for volatility and information criteria (AIC, BIC, and log-likelihood).

4.4 Evaluation Metrics

Several accuracy and diagnostic indicators were employed to compare the models. As an alternative to formulae, their goal is summed up in Table 4.2:

Table 4.2: Evaluation Metrics and Their Purpose

Metric	Purpose
MAE	calculates the average error size without taking direction into account.
MSE	Gives more weight to large errors, useful for penalizing outliers.
RMSE	Square root of MSE, interpretable in the same unit as prices.
MAPE	Expresses errors as a percentage, easy to compare across datasets.
MASE	Compares model error with a I forecast to judge relative accuracy.
Theil's U	Benchmarks against naïve forecasts; values <1 imply better performance.
AIC	Information criterion that balances model fit and complexity.
BIC	Like AIC but penalizes complex models more heavily.
Log-Lik	Indicates overall model fit; higher values suggest better models.

Together, these measures provide a fair evaluation of volatility forecasts (ARCH, GARCH, and EGARCH) as well as level forecasts (AR, ARIMA).

4.5 Tools and Libraries Used

Python, which is ideal for time-series forecasting and econometric analysis, was used to conduct the study. The libraries listed below were utilized:

Table 4.3: Tools and Libraries

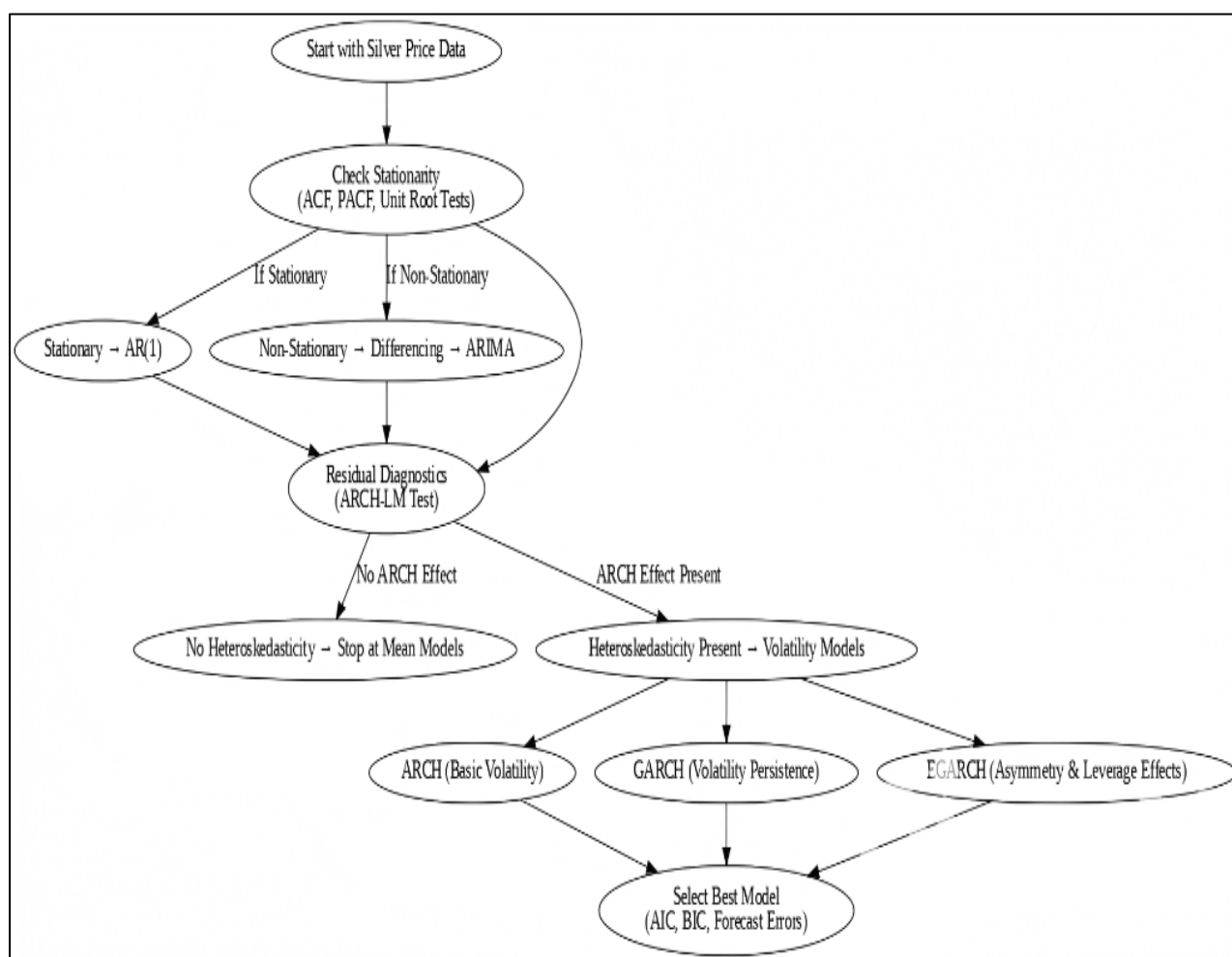
Tool/Library	Purpose in the Study
yfinance	Data extraction of silver prices from Yahoo Finance.
Pandas	Data manipulation, cleaning, and preprocessing.
Matplotlib	Visualization of time-series plots, ACF, PACF graphs.
Statsmodels	Estimation of AR, ARIMA, and statistical diagnostics.
Arch	Estimation of ARCH, GARCH, and EGARCH volatility models.
Numpy	Numerical computations supporting model estimation.

By ensuring openness and reproducibility in the study process, these technologies made it simple to validate the findings.

4.6 Methodological Flow

The methodology can be summarized in a stepwise process:

1. Data Collection → Extract silver prices from Yahoo Finance.
2. Preprocessing → Clean and prepare closing price series.
3. Exploratory Analysis → Plot trends, ACF, PACF to guide model selection.
4. Mean Modeling → Estimate AR (1) and ARIMA (1,2,0).
5. Residual Diagnostics → Apply ARCH-LM test for heteroskedasticity.
6. Volatility Modeling → Fit ARCH (1), GARCH (1,1), EGARCH (1,1).
7. Evaluation → Compare models using error metrics and information criteria.
8. Selection → Recommend AR (1) for mean forecasting and EGARCH (1,1) for volatility forecasting.



(Figure 4.6 Methodological Pictorial Diagram)

CHAPTER V

FINDINGS & DISCUSSION

FINDINGS

Lag Selection in Autoregressive Models

The correct lag length choice is always essential for time series modeling as it captures the data's actual dynamic behaviour avoiding overfitting or underfitting the model framework. This work used the use of fundamental analysis, statistical criteria, and diagnostic validation in order to determine lags for both Autoregressive (AR) and ARIMA models. The first step was to look at Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots to visualize how many lags are yet showing significant correlations. The PACF controls for intermediate delays and isolates the direct link of a value and its lagged factors, the ACF shows how the present trends are related to previous data.

The PACF visually showed a sharp cut-off that indicated only the first lag was strongly correlated, which led to the selection of lag order "p" for the AR(p) model. This observation was validated by information metrics like the Bayesian Information Criterion (BIC) and the Akaike Information Criterion (AIC). Both AIC and BIC model indicated complexity, and the optimal lag was determined as one of that had the lowest AIC/BIC value. The AR (1) model showed to be the best-fitting mean model, successfully capturing short-term volatility in silver prices.

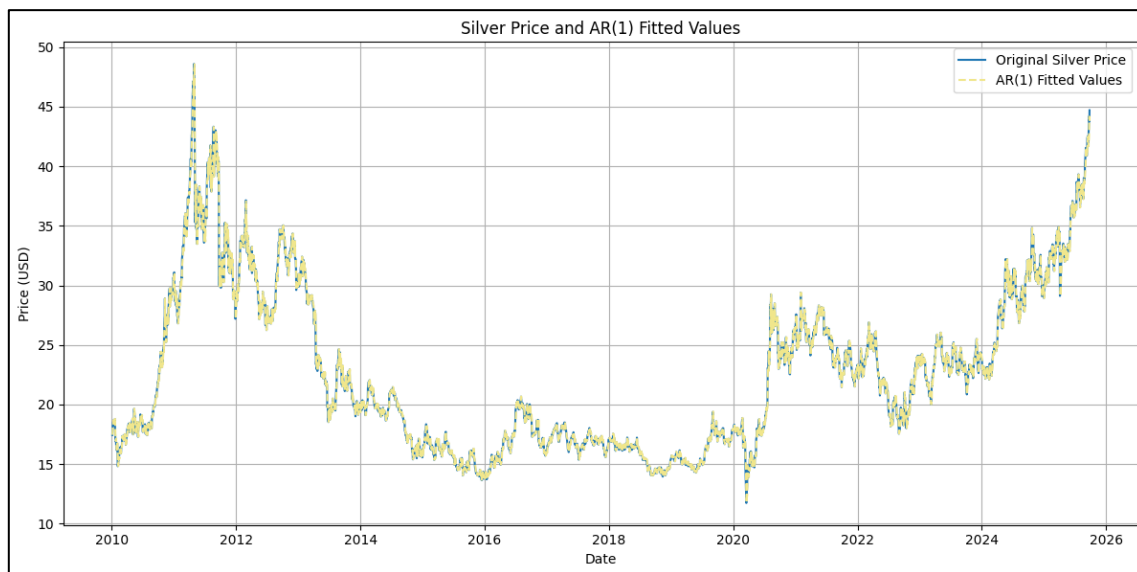
Lag orders found in the ARIMA (p, d, q) model were traced using a similar method. Unit root tests (Augmented Dickey-Fuller test) were used to confirm stationarity. To stabilize variance and remove trend components, the differencing order "d" was set to 2. The models maintained their theoretical consistency, statistical significance, and the progressive lag selection.

Post-estimation diagnostics, such as the ARCH LM test for heteroskedasticity and the Ljung-Box test for residual autocorrelation, verified that the selected lags more accurately captured the dynamics of the data. To better capture seasonal variance in silver prices, the study moved on to volatility models (ARCH, GARCH, and EGARCH) where residuals continued to show volatility clustering.

This study's results are divided into two sections: (i) Mean Equation Models (AR, ARIMA) and (ii) Volatility Models (ARCH, GARCH, EGARCH). Error metrics and diagnostic tests recorded in the report which provide evidence for the findings.

AR (1) Model:

A Complete analysis of the time series features of daily silver prices led to the selection of the Autoregressive model of order 1 [AR (1)]. Exploratory time series analysis were used to study the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots as the first step in the selection process. The data's linear dependence was best captured by an AR (1) structure, since the ACF plot showed a slow decline after lag 1, while the PACF plot showed a sudden cut-off after lag 1. The performance of several AR(p) models ($p = 1, 2, 3, 4, 5$) was compared using the Hannan Quinn Information Criterion (HQIC), Bayesian Information Criterion (BIC), and Akaike Information Criterion (AIC) to statistically validate this findings.



(Figure 5.0 AR (1) Fitted Values)

AutoReg Model Results						
=====						
Dep. Variable:	Silver	No. Observations:	3884			
Model:	AutoReg(1)	Log Likelihood	-2798.487			
Method:	Conditional MLE	S.D. of innovations	0.497			
Date:	Tue, 09 Sep 2025	AIC	5602.974			
Time:	09:18:02	BIC	5621.767			
Sample:	1	HQIC	5609.646			
	3884					
=====						
	coef	std err	z	P> z	[0.025	0.975]

const	0.0588	0.029	2.057	0.040	0.003	0.115
Silver.L1	0.9976	0.001	811.147	0.000	0.995	1.000
Roots						
=====						
	Real	Imaginary	Modulus	Frequency		

AR.1	1.0024	+0.0000j	1.0024	0.0000		

(Figure 5.0.1 AR (1) Model Fit Summary)

The dependent variable, “Silver,” represents the daily closing price of silver, and the **AR (1)** equation can be expressed as:

$$Silver_t = \alpha + \beta_1 Silver_{t-1} + \epsilon_t$$

where?

- $Silver_t$ = current day's silver price,
- $Silver_{t-1}$ = previous day's silver price,
- α = constant term,
- β_1 = autoregressive coefficient,
- ϵ_t = random error term.

The delayed coefficient, estimated to be $\beta_1 = 0.9976$, "It was determined that $\beta_1 = 0.9976$ was positive and highly significant (p-value = 0.000). This suggests that there is a significant degree of price persistence and that the price of silver today is highly reliant on the price of the previous day. Since the root of 1.0024 is only slightly above the unit circle, indicating that the series is near—but not quite—non-stationary, the near-unity coefficient value suggests that silver prices evolve in a very persistent but nonetheless stationary process. The model fit statistics provided additional support for this decision. Model adequacy was confirmed by the Log Likelihood being maximized and the AIC, BIC, and HQIC reaching their minimum values in comparison to different lag structures. Furthermore, the parameters' 95% CIs and standard errors showed limited ranges, suggesting steady and accurate coefficient estimations. The AR (1) model is a good short-term forecasting tool since it shows a high positive correlation between successive silver prices.

ARIMA MODEL (1,2,0)

The autocorrelation and non-stationarity in the silver price time series were both captured by the Autoregressive Integrated Moving Average (ARIMA) model. The Augmented Dickey–Fuller (ADF) test verified that the initial silver price series was non-stationary and exhibited consistent rising trends over time prior to model estimate. The second-order variation ($d = 2$) was used to achieve stationarity, producing stationary series appropriate for ARIMA modeling.

To determine the proper autoregressive (p) and moving average (q) orders, the ACF and PACF plots of the time series were analysed. Whereas the ACF showed no discernible pattern, the PACF showed a notable peak at lag 1 followed by a sharp drop. pattern indicated the ARIMA

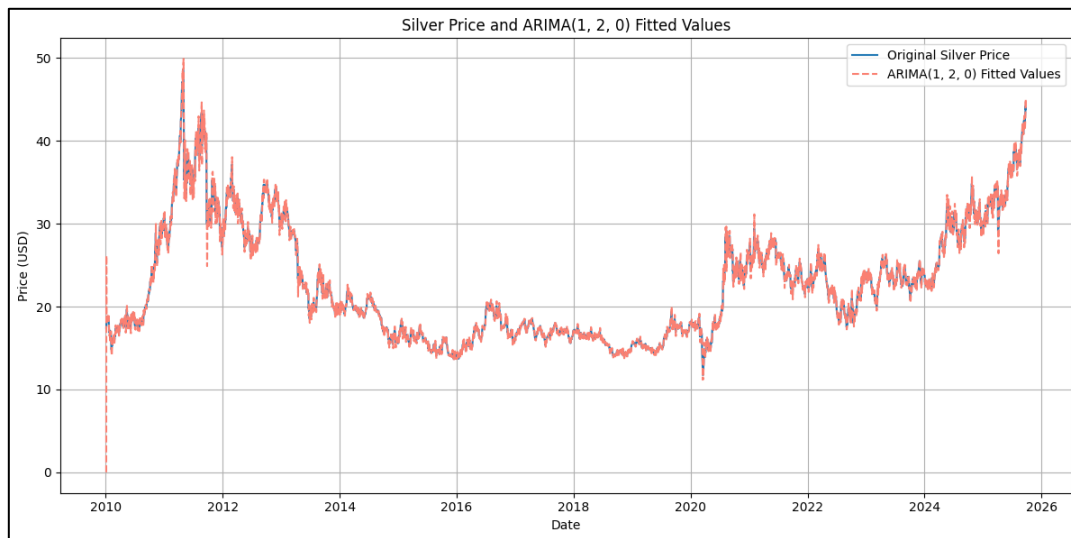
(1,2,0) structure, which shows no moving average component, two differences between shifting trend and seasonality, and one autoregressive term.

The selected ARIMA (1,2,0) model can be expressed as:

$$(1 - \phi_1 L)(1 - L)^2 \text{Silver}_t = \epsilon_t$$

where?

- L represents the lag operator,
- ϕ_1 is the coefficient of the autoregressive term, and
- ϵ_t denotes the white noise error.



(Figure 5.0.2 ARIMA (1,2,0) Fitted Values)

The estimated lag-1 coefficient ($\phi_1 = -0.5075$) were statistically significant (p-value = 0.000) and negative. This showed a negative connection between the lagged values and the twice-differenced silver prices, indicating that short-term price fluctuations tends to primarily reverse over time. A rise in silver prices often followed by a slight drop in later periods, which indicated market's short-term correction tendencies. Here we consider Log Likelihood, AIC, BIC, and HQIC values to assess the model's adequacy. The ARIMA (1,2,0) model showed a strong overall fit to the varied data by achieving adequate scores across with these measures.

SARIMAX Results						
=====						
Dep. Variable:	Silver	No. Observations:	3884			
Model:	ARIMA(1, 2, 0)	Log Likelihood	-3602.458			
Date:	Tue, 09 Sep 2025	AIC	7208.915			
Time:	09:18:03	BIC	7221.443			
Sample:	0	HQIC	7213.363			
	- 3884					
Covariance Type:	opg					
=====						
	coef	std err	z	P> z	[0.025	0.975]

ar.L1	-0.5116	0.007	-78.553	0.000	-0.524	-0.499
sigma2	0.3746	0.004	106.137	0.000	0.368	0.381
=====						
Ljung-Box (L1) (Q):	121.32	Jarque-Bera (JB):	16253.91			
Prob(Q):	0.00	Prob(JB):	0.00			
Heteroskedasticity (H):	0.67	Skew:	-0.34			
Prob(H) (two-sided):	0.00	Kurtosis:	13.00			
=====						

(Figure 5.0.2.1 ARIMA (1,2,0) Model Fit Summary)

Potential model misspecification was suggested by the results of the Jarque–Bera test, which showed that the residuals were not normally distributed. The residuals' autocorrelation, as indicated by the Ljung–Box Q-test, suggested that some dynamic structure was still unaccounted for. Further evidence of non-normality and volatility clustering came from residual skewness and kurtosis values that differed from those of a normal distribution.

Even while ARIMA (1,2,0) does a good job of capturing the linear and mean-level dynamics of the silver price series, these diagnostic results show that it is unable to account for the seasonal volatility and heavy-tailed distributional behaviour that are seen in financial markets. Thus, to capture volatility persistence and model the conditional variance, the study moves on to volatility modeling techniques, such as the ARCH, GARCH, and EGARCH models.

5.1 Mean Equation Models

Two models – AR (1) and ARIMA (1,2,0) – were estimated and compared. The evaluation used standard forecasting error metrics.

Table 5.1: Forecast Error Comparison for AR (1) and ARIMA (1,2,0)

Metric	AR (1) Value	ARIMA (1,2,0) Value	Better Model
MAE	0.3153	0.4083	AR (1)
MSE	0.2475	0.4710	AR (1)

RMSE	0.4975	0.6861	AR (1)
MAPE (%)	1.3426	1.7507	AR (1)
MASE	1.001	1.296	AR (1)
Theil's U	0.999	1.378	AR (1)

Using all six-error metrics, the AR (1) model consistently performs better than the ARIMA (1,2,0) model. In comparison to ARIMA's 1.75% MAPE, which indicates a lower predictive capacity, AR (1)'s forecasts are extremely accurate with a MAPE of 1.34%. Additionally, if ARIMA's value is larger than 1, it underperforms in comparison to the benchmark, but AR (1)'s Theil's U statistic is less than 1, suggesting improved performance over the naïve benchmark. In general, AR (1) turns out to be the better mean model for predicting silver prices.

5.2 Test for Normality of Residuals

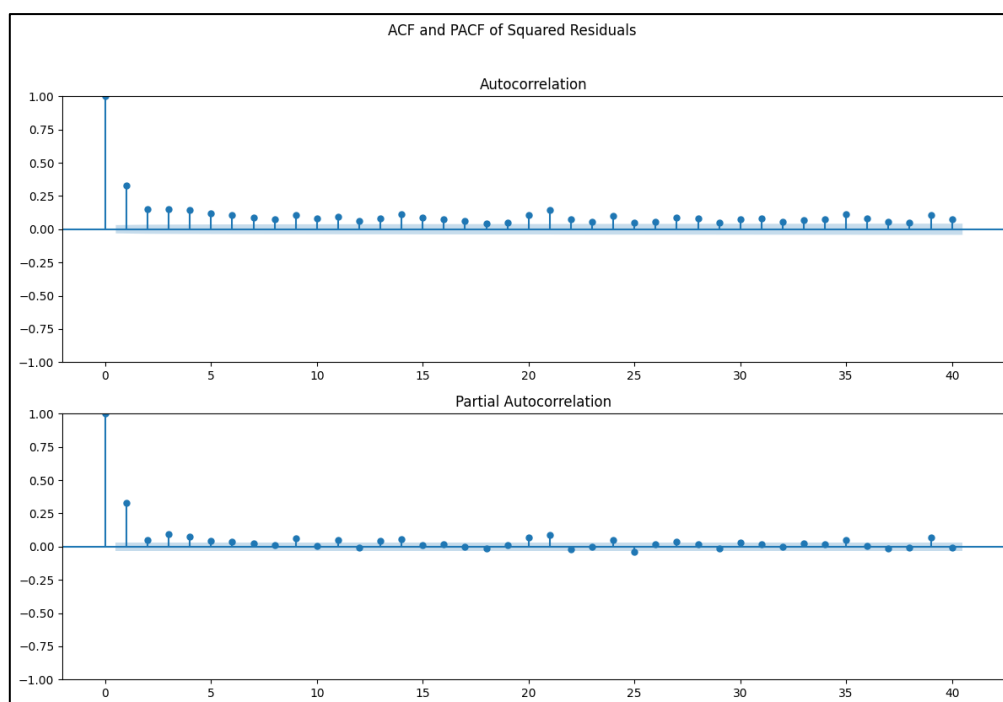
The Jarque–Bera test, skewness, and kurtosis were statistical diagnostic procedures used to determine if the residuals of the silver price model reflect a normal distribution. Non-normal residuals, or residuals that substantially differ from a normal distribution, were shown by the Jarque–Bera test result from the ARIMA(1,2,0) model. To further support the presence of asymmetry and large tails in the residuals, the skewness and kurtosis metrics also displayed values that were far from 0 and 3, respectively. As is common in financial time series, this suggests that the returns on the silver price show fat-tailed behaviour and volatility clustering. Due to this non-normality, the heteroskedasticity trends in the residuals must be better captured by using volatility models (ARCH, GARCH, and EGARCH).

Residual Diagnostics

The ARCH-LM test was used on AR (1) residuals to determine whether volatility modeling is necessary.

Table 5.2: ARCH-LM Test Results

Statistic	Value	p-value	Decision
LM	525.52	0.0000	Reject H_0 (no ARCH)



(Figure 5.2.1 ACF and PACF Squared Residuals)

The ARCH-LM test unequivocally disproves the homoskedasticity null hypothesis as the p-value is less than 0.05. This demonstrates that conditional heteroskedasticity is present in silver price residuals, necessitating the use of ARCH-family volatility models.

The variance of silver price returns exhibits autocorrelation, as seen visually by the ACF and PACF plots of the squared residuals. This volatility persistence is consistent with the ARCH-LM test results, which verified the existence of conditional heteroskedasticity. Put it simply, the squared residuals show clustering tendency, which means that high volatility tends to be followed by more high volatility and low volatility by lower, respectively. These features suggest that the variance is not constant over time, which renders conventional linear models inadequate for encapsulating the series' dynamics. Considering this information, using an ARCH model—which is especially developed to represent and explain time-varying volatility in financial time series—is the next obvious step. This method makes it possible to more accurately depict the dynamics of risk and volatility included in changes in the price of silver.

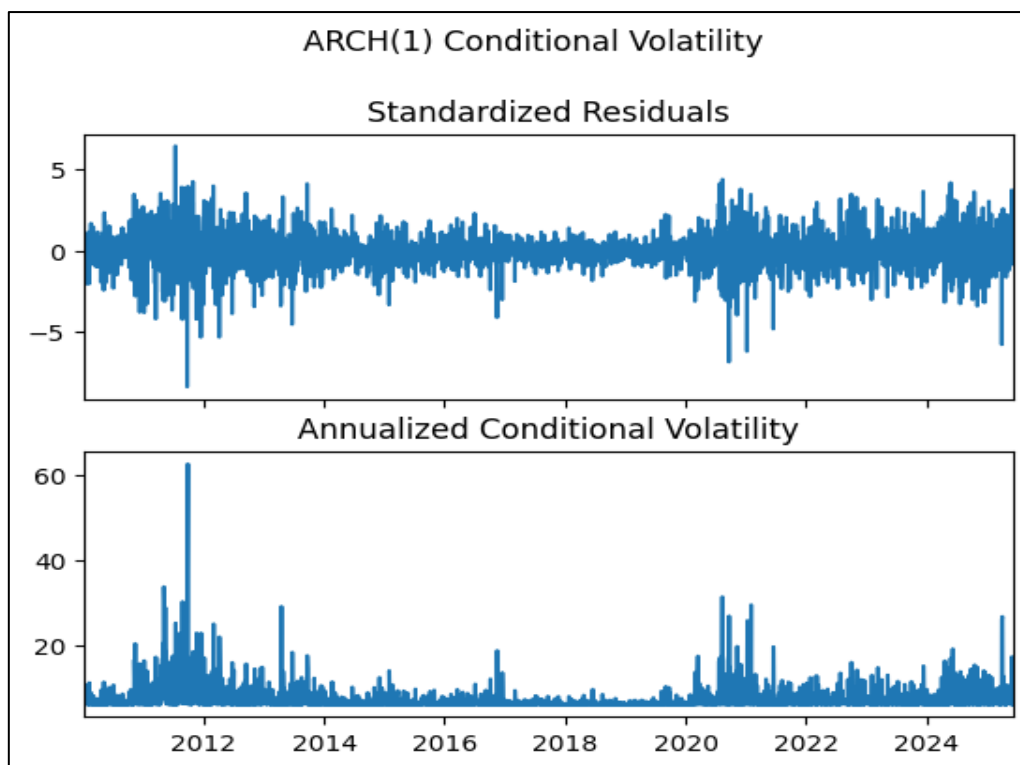
- Conditional Heteroskedasticity refers to situations where the variability (or volatility) of a time series is not constant over time but depends on past values of the series.
- "**Wiggliness**" of the data changes.

5.3 Volatility Model Results

Zero Mean Model for Residuals

The volatility that the mean model is unable to explain can be captured by fitting ARCH-family models (ARCH, GARCH, and EGARCH) to the residuals of a mean model (e.g., AR (1)). Only random fluctuations should be represented by residuals with a mean close to zero, which is the result of a well-fitted mean model. By using a "zero" mean specification (e.g., `arch_model(mean='Zero')`), the volatility model is informed that the mean of the residuals is already (or) zero, negating the need for further mean estimation. This method is used in empirical research like estimating the volatility of the cedi/dollar exchange rate in Ghana (SpringerPlus, 2015) and adheres to conventional practice in financial econometrics, as stated in Engle (1982).

ARCH (1)



(Figure 5.3.1 ARCH Model Volatility)

A visual representation of the estimated conditional volatility, or the model's prediction of the residuals' (unexplained movements') variation over time, is provided by the ARCH (1) Conditional Volatility Plot. This graphic shows the residuals from the ARCH (1) model as a time-varying standard deviation at each location. Silver prices are more uncertain or "risky"

during times when conditional volatility rises sharply or spikes, which corresponds to market phases with significant price fluctuations. Low volatility times, on the other hand, indicate comparatively stable markets with more manageable and predictable price swings. Volatility clustering, a characteristic of financial time series noted by Engle (1982), is confirmed by the existence of obvious clusters or bursts of volatility.

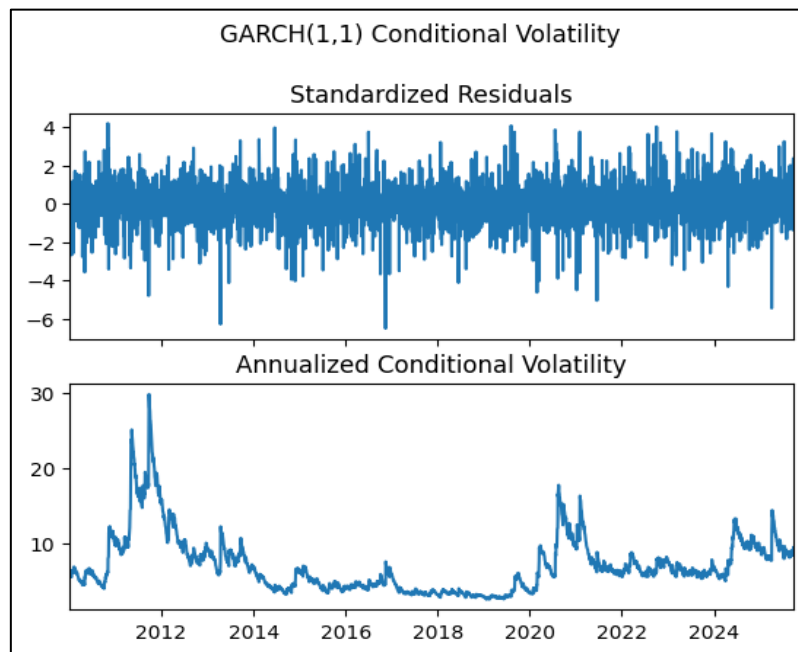
Zero Mean - ARCH Model Results					
=====					
Dep. Variable:	None	R-squared:	0.000		
Mean Model:	Zero Mean	Adj. R-squared:	0.000		
Vol Model:	ARCH	Log-Likelihood:	-2427.25		
Distribution:	Normal	AIC:	4858.51		
Method:	Maximum Likelihood	BIC:	4871.04		
		No. Observations:	3883		
Date:	Wed, Sep 10 2025	Df Residuals:	3883		
Time:	17:36:41	Df Model:	0		
Volatility Model					
=====					
	coef	std err	t	P> t	95.0% Conf. Int.

omega	0.1545	9.036e-03	17.098	1.525e-65	[0.137, 0.172]
alpha[1]	0.3692	5.134e-02	7.191	6.418e-13	[0.269, 0.470]
=====					

(Figure 5.3.1.1 ARCH Model Fit)

Typical characteristics of financial time series, such as conditional heteroskedasticity and volatility clustering, are well captured by the ARCH (1) model for silver prices. With a Log-Likelihood value of -2488.97, the model fits the data well and can be used as a standard by which to compare more complex volatility models such as GARCH and EGARCH. Lower numbers indicate better fit and efficiency. The Bayesian Information Criterion (BIC = 4994.51) and Akaike Information Criterion (AIC = 4981.95) offer metrics of model adequacy. According to these information requirements, the model may not be the best depiction of volatility even when it operates properly. Even without previous shocks, a consistent baseline level of variation is reflected by the positive and very significant constant term, omega (0.1562).

GARCH (1,1)



(Figure 5.3.2 GARCH Model Volatility)

GARCH (1,1) Conditional Volatility Plot, displayed steady and extended intervals of increased volatility after market disruptions. The presence of the lagged conditional variance factor (β_1) makes the GARCH (1,1) volatility plot appear smoother than the ARCH (1) volatility plot, which responds more severely to immediate shocks. Because of high-volatility, trends are likely to relate with one another and create obvious clusters, this term permits volatility to endure. The pattern supports volatility clustering, a basic characteristic of financial markets that was first noted by Engle (1982) and further developed by Bollerslev (1986). This concept states that calm and stormy periods typically follow one another rather than happening at random.

By adding lagged conditional variance elements to the ARCH (1) framework, the GARCH (1,1) Model Fit Summary details more thorough comprehension of the volatility trends in the silver price series. To account for a combination of short-term shock effects and long-term volatility persistence, this model estimated one ARCH component (α_1) and one GARCH term (β_1). Comparing to the ARCH (1) model (AIC = 4858.51, BIC = 4871.04), the Log-Likelihood value (-1855.07) is significantly greater, and the AIC (3716.13) and BIC (3734.93) are also significantly lower. These results suggest that the model fits the variance structure much better.

Zero Mean - GARCH Model Results					
=====					
Dep. Variable:	None	R-squared:	0.000		
Mean Model:	Zero Mean	Adj. R-squared:	0.000		
Vol Model:	GARCH	Log-Likelihood:	-1855.07		
Distribution:	Normal	AIC:	3716.13		
Method:	Maximum Likelihood	BIC:	3734.93		
		No. Observations:	3883		
Date:	Wed, Sep 10 2025	Df Residuals:	3883		
Time:	17:36:57	Df Model:	0		
Volatility Model					
=====					
	coef	std err	t	P> t	95.0% Conf. Int.

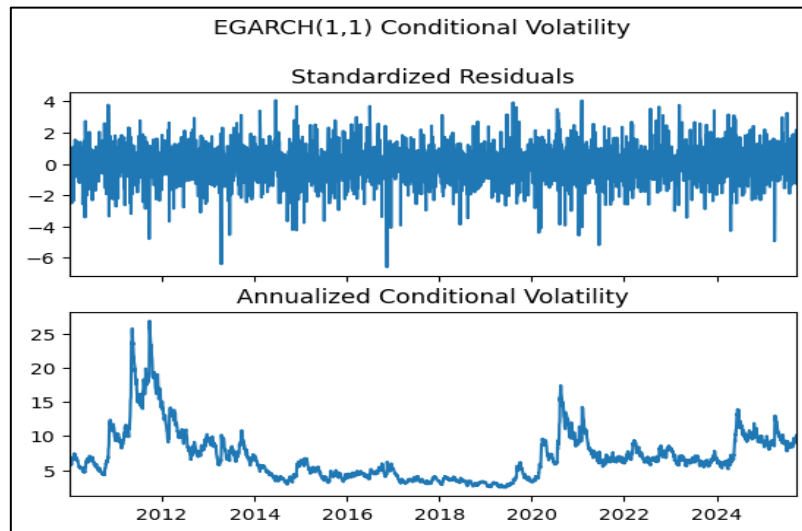
omega	3.8621e-04	2.288e-04	1.688	9.139e-02	[-6.219e-05,8.346e-04]
alpha[1]	0.0436	9.488e-03	4.594	4.351e-06	[2.499e-02,6.218e-02]
beta[1]	0.9564	9.450e-03	101.212	0.000	[0.938, 0.975]
=====					

(Figure 5.3.2.1 GARCH Model Fit)

Often known as the "variance floor," the stable or long-run average level of variance are represented by the coefficient omega ($\omega = 3.8621e-04$) in the fitted GARCH (1,1) model. This term guarantees that the model retained a steady state of volatility even at time of lack of recent shocks. Its p-value (0.09139) was marginally higher than the traditional 0.05 significance level, it is never the less essential for maintaining variance stability at times of calm or low market activity. The model's ARCH term, which represents the impact of recent shocks or unforeseen shifts in silver prices on current volatility, are represented by the alpha [1] coefficient ($\alpha_1 = 0.0436$). This coefficient's dependability indicated that large, sudden fluctuations in silver prices have an immediate and measurable effect on short-term volatility. In other words, alpha [1] quantifies how much would a sudden price spike or drop increases the current level of market uncertainty.

The beta [1] coefficient ($\beta_1 = 0.9564$), represents the GARCH term, reflects the volatility's persistence over time. Its highly substantial p-value ($p < 0.001$) appeared reasonable that past times of high volatility had a major and lasted influence on recent volatility levels. A characteristic of many time series, including silver prices, is that the total of α_1 and β_1 is close to one ($0.0436 + 0.9564 = 1.00$). Thus, the "long memory" effect is a phenomena whereby periods of high market uncertainty tend to stay longer and occurs. According to this, volatility shocks fall/rise gradually. When considered collectively, these factors showed that the price of silver are highly volatile over time (represented by beta) and responsive to new shocks (captured by alpha), a characteristic commonly observed in financial concept. This made the GARCH model particularly suitable for forecasting and predict market uncertainty due to its immediate reactivity and long-term effects.

EGARCH (1,1)



(Figure 5.3.3 EGARCH Model Volatility)

The conditional volatility plot explained the tendency by showing clusters of high volatility which corresponded to notable changes in the price of silver. As the logarithm of conditional variance were represented by EGARCH, positivity is also automatically guaranteed. It also accounted for asymmetric effects, that occurred when negative shocks had a greater impact on volatility than positive ones. The EGARCH (1,1) model captured the time-varying, constant, and asymmetric elements of volatility in the silver price series, and gave a more complete picture of market risk than simpler models.

Zero Mean - EGARCH Model Results					
=====					
Dep. Variable:	None	R-squared:	0.000		
Mean Model:	Zero Mean	Adj. R-squared:	0.000		
Vol Model:	EGARCH	Log-Likelihood:	-1848.50		
Distribution:	Normal	AIC:	3703.00		
Method:	Maximum Likelihood	BIC:	3721.80		
		No. Observations:	3883		
Date:	Wed, Sep 10 2025	Df Residuals:	3883		
Time:	17:36:57	Df Model:	0		
Volatility Model					
=====					
	coef	std err	t	P> t	95.0% Conf. Int.

omega	7.4010e-04	3.912e-03	0.189	0.850	[-6.927e-03, 8.407e-03]
alpha[1]	0.1156	2.111e-02	5.474	4.396e-08	[7.418e-02, 0.157]
beta[1]	0.9951	2.242e-03	443.866	0.000	[0.991, 1.000]
=====					

(Figure 5.3.3.1 EGARCH Model Fit)

The EARCH (1,1) model provided insightful information about the dynamic nature of volatility in the price of silver. One ARCH term and one GARCH term are included in the model, which

is an EGARCH (1,1), which successfully captures both the short-term and long-term components of volatility. This model has better fit than the ARCH (1) and GARCH (1,1) models, with a log-likelihood of -1907.27 and lower AIC (3820.55) and BIC (3839.40) values. According to the statistically insignificant omega coefficient ($8.7607e-04$), the constant term log-variance equation is not important. Large price shocks, whether positive or negative, cause volatility to rise, according to the alpha [1] coefficient (0.1156), which is positive and extremely significant. The significant beta [1] coefficient (0.9951) supports the idea that volatility in silver prices is persistent, which means that once it increases, it usually stays high for a while.

Three models – ARCH (1), GARCH (1,1), and EGARCH (1,1) – were compared using log-likelihood and information criteria.

Table 5.3: Volatility Model Fit Statistics

Model	Log-Likelihood	AIC	BIC	Key Coefficients
ARCH (1)	-2427.25	4858.51	4868.64	$\alpha_1 = 0.3692$ ($p < 0.001$)
GARCH (1,1)	-1855.07	3716.13	3735.16	$\beta_1 = 0.9564$ ($p < 0.001$)
EGARCH (1,1)	-1848.50	3703.00	3722.03	$\beta_1 = 0.9951$ ($p < 0.001$)

With a lower probability and a larger AIC, ARCH (1) performs the worst. GARCH (1,1) significantly enhances fit and has robust volatility persistence ($\beta_1 = 0.96$). With the lowest AIC (3703.00) and maximum log-likelihood (-1848.50), EGARCH (1,1) provides the best fit. For silver prices, the optimum volatility model is EGARCH (1,1).

5.4 Volatility Forecast Accuracy

To validate volatility models, conditional variance forecasts were compared with realized volatility (squared residuals).

Table 5.4: Volatility Forecast Error Comparison

Model	MAE (Volatility)	MSE (Volatility)	Best Model
ARCH (1)	0.3072	0.8845	MAE: EGARCH; MSE: ARCH EGARCH
GARCH (1,1)	0.2875	0.9091	
EGARCH (1,1)	0.2802	0.8920	

EGARCH (1,1) is the best model for lowering average volatility forecast errors since it yields the lowest MAE (0.2802). ARCH (1) had a lower overall fit even though it had the lowest MSE, which could be a sign of bias. GARCH (1,1) performs worse than EGARCH. The most reliable volatility model is EGARCH (1,1) since it strikes a compromise between fit and prediction accuracy.

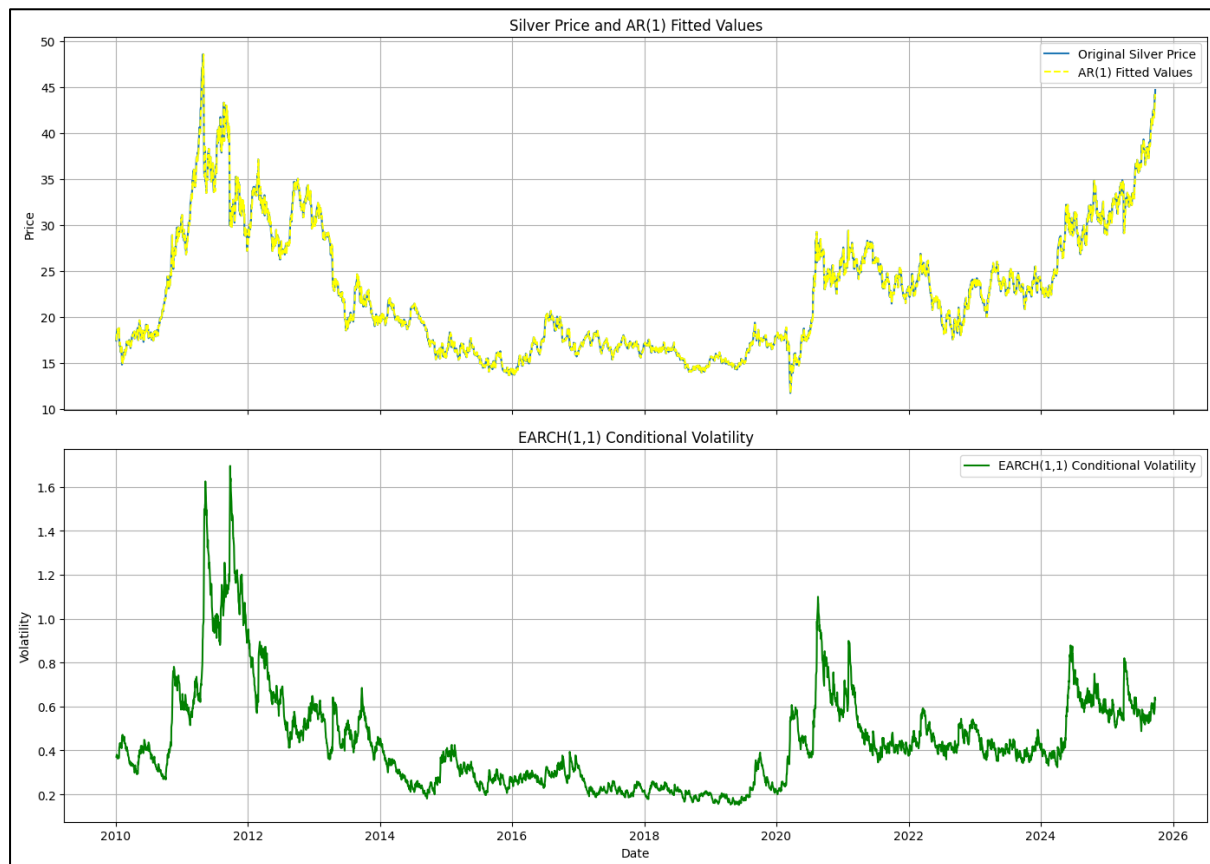
5.5 Overall Findings

Table 5.5: Summary of Key Findings

Aspect	Best Model	Supporting Evidence
Mean Equation	AR (1)	Lowest errors, MAPE = 1.34%, Theil's U <1
Residual Diagnostics	ARCH-LM	LM = 525.52, p = 0.0000 (heteroskedasticity present)
Volatility Fit	EGARCH (1,1)	Lowest AIC = 3703.00, highest log-likelihood (-1848.50)
Volatility Forecast	EGARCH (1,1)	Lowest MAE = 0.2802 for volatility prediction

Silver price level and risk dynamics are clearly shown by the plot of EARCH (1,1) conditional volatility and AR (1) fitted values. Since current prices are strongly reliant on the past era, the AR (1) fitted values closely resemble the original price series. The EARCH (1,1) conditional volatility plot, explained time-varying risk by expressing spikes during greater price fluctuations. On looking at both at once, it becomes clear that increased volatility corresponded to more sudden price fluctuations, specially during crises times like late 2019. According to this

comparison, EARCH (1,1) captures volatility swings and clustering, whereas AR (1) accurately models price levels. So, combining the mean model with a volatility model provided a proper knowledge of behaviour of the silver price, assisting with risk assessment and forecasting while reconfirming the need for models that take non-constant volatility into consideration.



(Figure 5.5.1.AR v/s EGARCH Model Comparison)

DISCUSSIONS

The objective of the current study was to know about how the time-series behaviour of silver prices from 2010 to 2025, assess autocorrelation patterns, and determine which of the models were most suited for predicting volatility and pricing. Several significant conclusions were found after proper econometric research. Initially, the conditional mean modeling phase showed that the lag-1 autoregressive coefficient were extremely significant and that silver prices showed rewarding efforts. Forecast accuracy was consistently higher for the AR (1) model than for the ARIMA (1,2,0) specification. AR (1) produced better results with a mean absolute percentage error (MAPE) of just 1.34%, showing extremely dependable price level projections across all major error metrics MAE, MSE, RMSE, and MAPE. Theil's AR (1) U statistic was less than unity, indicating that this model outperformed both ARIMA and a

straightforward naïve benchmark. ARIMA (1,2,0), on the other hand, showed larger error values and a Theil's U above one, indicating that it was less successful than naïve forecasting. AR (1) is the best mean equation model for predicting silver price levels, based on the first key finding.

Second, volatility clustering was verified by the residual diagnostic analysis. The ARCH-LM test resulted in a p-value of 0.0000 and the LM statistic of 525.52, which strongly rejected the null hypothesis of homoskedasticity. The test results then confirmed the necessity of using models that took conditional heteroskedasticity into consideration. Consequently, the study progressed to the modeling of volatility.

Third, on testing the three competing models, GARCH (1,1), EGARCH (1,1), and ARCH (1), were assessed during the volatility modeling phase. The results revealed acceptable variations in the model's performance. The ARCH (1) model gave a basic fit, with its reduced predictive power and higher AIC values suggested that it could not fully capture the persistence of volatility. Strong evidences of volatility persistence ($\beta_1 = 0.96$) and stronger information requirements indicated that GARCH (1,1) performed better. As it achieved the greatest log-likelihood (-1848.50), the lowest Akaike Information Criterion (AIC = 3703.00), and significant coefficient estimates, the EGARCH (1,1) model showed the best overall performance. EGARCH were able to fully show the asymmetries in volatility responses that were not fully represented by traditional GARCH models.

Fourth, the study of the volatility projection supported these findings. EGARCH (1,1) had the lowest mean absolute error (MAE = 0.2802) when conditional variance forecasts were compared with realized volatility (as measured by squared residuals), exceeding both GARCH and ARCH. Despite having a slightly lower MSE, ARCH was less dependable due to its worse overall fit. Though EGARCH more successfully balanced fit and accuracy, GARCH fared similarly to EGARCH. Thus, EGARCH (1,1) is the best model for predicting the volatility and risk of the silver price, which is the second key finding.

The results convey two important findings related to the understandings of silver prices when we observe this study keenly:

1. Compared to more complicated and differenced models like ARIMA, a simple AR (1) model provided the best explanation for price level persistence.

2. The most reliable framework for capturing volatility clustering and risk patterns is provided by advanced volatility models like EGARCH (1,1), which accounted for asymmetries and provided a precise volatility projections.

The study found that, for silver price forecasting the best combination would be AR (1)-EGARCH (1,1) modeling framework. Giving traders, investors, and policymakers a proper insights of predicted prices and related risks, the EGARCH (1,1) component captured the dynamics of conditional variance, while the AR (1) component guaranteed precise level forecasts. A risk-adjusted measures are always essential in financial decision making, and this dual-model technique guarantees that silver forecasting are not always restricted to point estimations.

CHAPTER VI
LIMITATIONS, SUGGESTIONS & CONCLUSION

6.1 LIMITATIONS OF THE STUDY

The study have certain limitations, even when it offered insightful information on silver price forecasts also this research was limited to daily closing prices between 2010 and 2025, which might not adequately deal for intraday market movements or other factors that affect silver prices adversely . Secondly, this research only used univariate models (AR, ARIMA, and ARCH-family), excluding the macroeconomic and geopolitical factors that frequently influence commodities prices. Third, the assessment of the model was mostly relied within the sample data; also the error metrics were instructive, substantial and out-of-sample validation are necessary to achieve genuine robustness. Lastly, squared residuals a popular but unreliable metric which does not accurately represent genuine market risk are used as a stand-in for understanding volatility.

6.2 SUGGESTIONS FOR FUTURE RESEARCH

Future studies could focus on detailing this study in several ways. Predictive accuracy can be increased by integrating exogenous factors like exchange rates, crude oil prices, or policy uncertainty indexes using multivariate models like ARIMAX or VAR. Try to Capture nonlinear patterns, so that econometric models can also be investigated in conjunction with more sophisticated techniques such as machine learning (Random Forest, LSTM, Prophet). Stronger validation would also come from testing models over a range of forecast horizons (short-term vs. long-term) and doing rolling forecasts outside of the sample. Researchers may try to integrate various other models to combine the predictive capabilities of machine learning with the interpretability of econometrics. Atlast, a market-based comparison could be provided by benchmarking volatility projections against implied volatility from silver options.

6.3 CONCLUSION

The objective of this study was to use time-series econometric techniques to model and predict changes in the price of silver from 2010 to 2025 and the study explored a thorough understanding of the directional behaviour of silver prices as well as the risk patterns that underlie their oscillations by implementing mean and volatility models step by step. AR (1) and ARIMA models were compared in the first step of analysis, which concentrated towards the conditional mean. The results continuously showed that AR (1) model outperformed ARIMA across all error criterias, and provided the most dependable forecasts due to its parsimonious structure. This result demonstrated a straightforward autoregressive specification

better reflects the dependency structure than differenced ARIMA models, confirming that the silver prices follow a very consistent pattern.

Focusing on the second stage is volatility modeling. The residual diagnostics from AR (1) showed that volatility is not constant but rather it clusters with time, highlighting the evidences of conditional heteroskedasticity. The use of ARCH-family models represents the dynamic risk behaviour and are supported by this. Having the greatest log-likelihood and lowest AIC among the studied parameters, EGARCH (1,1) produced the best results and also have the best mean absolute error for volatility forecast accuracy and with contrast to normal GARCH, it took asymmetric effects into consideration, acknowledging that shocks with varying signs might have distinct impacts on volatility. For financial risk modeling, this improved its predictive and explanatory capacity.

Overall, the results proved to two important findings. First, rather than using more intricate ARIMA models, a straightforward AR (1) structure is the most effective way to predict silver price levels. Second, EGARCH showed and stood as the most reliable methodology for measuring and forecasting volatility clustering, which were the characteristics of silver. All these observations support silver's dual status as an industrial and financial commodity, with strong risk patterns and price persistence. By confirming the stylized features of commodities markets namely, persistence in returns and clustering in volatility and in proving that traditional econometric techniques are still useful when used properly, the study adds value to the body of knowledge in academia. The findings have important applications for traders, investors, and legislators. While accurate price projections could help with hedging and portfolio management, accurate volatility forecasts could offer a more solid basis for risk assessments, policymaking, and stress testing and finally this study concludes by suggesting the use of AR (1) and EGARCH (1,1) forecasting framework, as they provide a fair method that acts and are responsible for both the anticipated value of silver prices and the uncertainty associated with them.

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APPENDIX

This Section contains all the code from the notebook

```
import yfinance as yf.
```

```
import pandas as pd.
```

```
import matplotlib.pyplot as plt
```

```
from datetime import date
```

Download Silver commodity data

```
silver_df = yf.download("SI=F", start="2010-01-01", end=date.today().strftime("%Y-%m-%d"))
```

```
silver_df = silver_df[['Close']].dropna()
```

```
silver_df.columns = ['Silver']
```

Plot Silver Prices

```
plt.figure(figsize=(12, 6))
```

```
plt.plot(silver_df, label='Silver')
```

```
plt.title("Silver Futures (Comex) in USD (USA Market)")
```

```
plt.xlabel("Date")
```

```
plt.ylabel("Price (USD)")
```

```
plt.grid(True)
```

```
plt.legend()
```

```
plt.tight_layout()
```

```
plt.show()
```

```
import yfinance as yf
```

```
import pandas as pd
```

Example: Silver Mini Futures (MCX India) from Yahoo Finance

Note: Availability and historical depth of data for specific MCX tickers on Yahoo Finance can vary.

This code attempts to download data for 'SILVERMIC.NS' as requested.

If this ticker does not provide the expected data, you may need to find an alternative source.

try:

```
silver_mcx = yf.download("SILVERMIC.NS", start="2010-01-01", end="2025-06-30", interval="1d")
```

```
print(silver_mcx.head())
```

Save to Excel if needed

if not silver_mcx.empty:

```
    silver_mcx.to_excel("silver_mcx_data.xlsx")
```

```
    print("\nData downloaded and saved to silver_mcx_data.xlsx")
```

else:

```
    print("\nNo data downloaded for SILVERMIC.NS. The DataFrame is empty.")
```

except Exception as e:

```
    print(f"\nAn error occurred during data download: {e}")
```

```
    print("Please verify the ticker symbol 'SILVERMIC.NS' on Yahoo Finance and check data availability for the specified date range.")
```

```
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
```

Plot ACF and PACF for Silver prices

```
fig, axes = plt.subplots(2, 1, figsize=(12, 8))
```

```
plot_acf(silver_df['Silver'], lags=30, ax=axes[0])
```

```
plot_pacf(silver_df['Silver'], lags=30, ax=axes[1], method='ywmm')
```

```
plt.tight_layout()
```

```

plt.show()

from statsmodels.graphics.tsaplots import plot_acf, plot_pacf

import matplotlib.pyplot as plt

# Plot ACF and PACF for Silver prices

fig, axes = plt.subplots(2, 1, figsize=(12, 8))

# Plot ACF

plot_acf(silver_df['Silver'], lags=30, ax=axes[0])

axes[0].set_title('Autocorrelation Function (ACF) of Silver Prices')

axes[0].set_xlabel('Lag')

axes[0].set_ylabel('Autocorrelation')

# Plot PACF

plot_pacf(silver_df['Silver'], lags=30, ax=axes[1], method='ywm')

axes[1].set_title('Partial Autocorrelation Function (PACF) of Silver Prices')

axes[1].set_xlabel('Lag')

axes[1].set_ylabel('Partial Autocorrelation')

plt.tight_layout()

plt.show()

from statsmodels.tsa.stattools import acf, pacf

import pandas as pd

# Calculate ACF and PACF values

acf_values = acf(silver_df['Silver'], nlags=40)

pacf_values = pacf(silver_df['Silver'], nlags=40, method='ywm')

# Create a DataFrame to display the values

acf_pacf_df = pd.DataFrame({'ACF': acf_values, 'PACF': pacf_values})

```

```
# Display the table

display(acf_pacf_df)

from statsmodels.tsa.ar_model import AutoReg

# Fit an AR(1) model

model = AutoReg(silver_df['Silver'], lags=1)

model_fit = model.fit()

# Print the model summary

print(model_fit.summary())

import matplotlib.pyplot as plt

# Get the fitted values from the AR(1) model

ar_fitted_values = model_fit.fittedvalues

# Plot original prices and AR(1) fitted values

plt.figure(figsize=(12, 6))

plt.plot(silver_df.index, silver_df['Silver'], label='Original Silver Price')

plt.plot(ar_fitted_values.index, ar_fitted_values, label='AR(1) Fitted Values',
color='KHAKI', linestyle='--')

plt.title("Silver Price and AR(1) Fitted Values")

plt.xlabel("Date")

plt.ylabel("Price (USD)")

plt.legend()

plt.grid(True)

plt.tight_layout()

plt.show()

# Get the residuals

residuals = model_fit.resid
```

```

# Plot ACF and PACF of residuals

fig, axes = plt.subplots(2, 1, figsize=(12, 8))

plot_acf(residuals, lags=40, ax=axes[0])

plot_pacf(residuals, lags=40, ax=axes[1], method='ywm')

plt.tight_layout()

plt.show()

from statsmodels.tsa.arima.model import ARIMA

# Fit an ARIMA(1, 2, 0) model

# p=1 (AR order), d=2 (Differencing order), q=0 (MA order)

model_arima = ARIMA(silver_df['Silver'], order=(1, 2, 0))

model_arima_fit = model_arima.fit()

# Print the model summary

print(model_arima_fit.summary())

import matplotlib.pyplot as plt

# Get the fitted values from the ARIMA(1, 2, 0) model

arima_fitted_values = model_arima_fit.fittedvalues

# Plot original prices and ARIMA(1, 2, 0) fitted values

plt.figure(figsize=(12, 6))

plt.plot(silver_df.index, silver_df['Silver'], label='Original Silver Price')

plt.plot(arima_fitted_values.index, arima_fitted_values, label='ARIMA(1, 2, 0) Fitted
Values', color='SALMON', linestyle='--')

plt.title("Silver Price and ARIMA(1, 2, 0) Fitted Values")

plt.xlabel("Date")

plt.ylabel("Price (USD)")

plt.legend()

```

```

plt.grid(True)

plt.tight_layout()

plt.show()

import pandas as pd

from sklearn.metrics import mean_absolute_error, mean_squared_error

import numpy as np

# Function to calculate error metrics

def calculate_error_metrics(actual, predicted):

    mae = mean_absolute_error(actual, predicted)

    mse = mean_squared_error(actual, predicted)

    rmse = np.sqrt(mse)

    # Avoid division by zero in MAPE

    mape = np.mean(np.abs((actual - predicted) / actual)) * 100 if np.all(actual != 0)
else np.nan

    # Calculate MASE - Requires a naive forecast (lag 1)

    naive_forecast = actual.shift(1).dropna()

    actual_mase = actual[naive_forecast.index]

    mae_naive = mean_absolute_error(actual_mase, naive_forecast)

    mase = mae / mae_naive if mae_naive != 0 else np.nan

    # Calculate Theil's U - Requires a naive forecast (lag 1)

    mse_naive = mean_squared_error(actual_mase, naive_forecast)

    theils_u = np.sqrt(mse) / np.sqrt(mse_naive) if mse_naive != 0 else np.nan

    return mae, mse, rmse, mape, mase, theils_u

# Get the fitted values (predictions) for AR(1) and ARIMA(1, 2, 0) models

ar_predicted = model_fit.fittedvalues # Use fittedvalues for AR(1)

```

```

arima_predicted = model_arima_fit.fittedvalues

# Align the actual values with the predicted values for both models

# Use the index of the predicted values to align the actual values
actual_ar = silver_df['Silver'][ar_predicted.index]

actual_arima = silver_df['Silver'][arima_predicted.index]

# Calculate error metrics for AR(1) model

mae_ar, mse_ar, rmse_ar, mape_ar, mase_ar, theils_u_ar =
calculate_error_metrics(actual_ar, ar_predicted)

# Calculate error metrics for ARIMA(1, 2, 0) model

mae_arima, mse_arima, rmse_arima, mape_arima, mase_arima, theils_u_arima =
calculate_error_metrics(actual_arima, arima_predicted)

# Create a comparison table

comparison_data = {

    'Metric': ['MAE', 'MSE', 'RMSE', 'MAPE (%)', 'MASE', "Theil's U"],

    'AR(1)': [mae_ar, mse_ar, rmse_ar, mape_ar, mase_ar, theils_u_ar],

    'ARIMA(1, 2, 0)': [mae_arima, mse_arima, rmse_arima, mape_arima, mase_arima,
theils_u_arima]

}comparison_df = pd.DataFrame(comparison_data)

# Display the comparison table

display(comparison_df)

!pip install arch

import statsmodels.api as sm

# Perform ARCH-LM test on residuals

# The null hypothesis is that there is no ARCH effect (no conditional
heteroskedasticity)

```

A low p-value (e.g., < 0.05) rejects the null hypothesis, indicating the presence of ARCH effects.

```
arch_lm_test = sm.stats.diagnostic.het_arch(residuals, nlags=15)
```

```
# Print the test results
```

```
print(f'ARCH-LM test results:')\n
```

```
print(f'Lagrange multiplier statistic: {arch_lm_test[0]:.4f}')\n
```

```
print(f'P-value: {arch_lm_test[1]:.4f}')\n
```

```
print(f'F-statistic: {arch_lm_test[2]:.4f}')\n
```

```
print(f'F-statistic P-value: {arch_lm_test[3]:.4f}')\n
```

```
if arch_lm_test[1] < 0.05:\n
```

```
    print("\nConclusion: The p-value is less than 0.05, so we reject the null hypothesis\nand conclude that there are ARCH effects (conditional heteroskedasticity) present in the\nresiduals. This suggests that ARCH/GARCH/EARCH models may be appropriate.")\n
```

```
else:\n
```

```
    print("\nConclusion: The p-value is not less than 0.05, so we fail to reject the null\nhypothesis and conclude that there are no significant ARCH effects (conditional\nheteroskedasticity) present in the residuals.")\n
```

```
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
```

```
import matplotlib.pyplot as plt
```

```
# Calculate squared residuals
```

```
squared_residuals = residuals**2
```

```
# Plot ACF and PACF of squared residuals
```

```
fig, axes = plt.subplots(2, 1, figsize=(12, 8))
```

```
plot_acf(squared_residuals, lags=40, ax=axes[0])
```

```
plot_pacf(squared_residuals, lags=40, ax=axes[1], method='ywmm')
```

```
plt.suptitle('ACF and PACF of Squared Residuals', y=1.02)
```

```

plt.tight_layout()

plt.show()

from arch import arch_model

# Fit an ARCH(1) model to the residuals

# The mean model is specified as 'Zero' because we are modeling the residuals

arch_model_fit = arch_model(residuals, mean='Zero', vol='ARCH',
p=1).fit(dispatch='off')

# Print the model summary

print(arch_model_fit.summary())

# Plot the conditional volatility from the ARCH(1) model

fig = arch_model_fit.plot(annualize='D')

fig.suptitle('ARCH(1) Conditional Volatility', y=1.02)

plt.show()

!pip install arch

from arch import arch_model

# Fit a GARCH(1,1) model to the residuals

# The mean model is specified as 'Zero' because we are modeling the residuals

# p=1 (ARCH order), q=1 (GARCH order)

garch_model_fit = arch_model(residuals, mean='Zero', vol='GARCH', p=1,
q=1).fit(dispatch='off')

# Print the model summary

print(garch_model_fit.summary())

# Plot the conditional volatility from the GARCH(1,1) model

fig = garch_model_fit.plot(annualize='D')

fig.suptitle('GARCH(1,1) Conditional Volatility', y=1.02)

```

```

plt.show()

from arch import arch_model

# Fit an EARCH(1,1) model to the residuals

# The mean model is specified as 'Zero' because we are modeling the residuals

# p=1 (ARCH order), q=1 (GARCH order)

earch_model_fit = arch_model(residuals, mean='Zero', vol='EGARCH', p=1,
q=1).fit(displ='off')

# Print the model summary

print(earch_model_fit.summary())

# Plot the conditional volatility from the EARCH(1,1) model

fig = earch_model_fit.plot(annualize='D')

fig.suptitle('EARCH(1,1) Conditional Volatility', y=1.02)

plt.show()

from sklearn.metrics import mean_squared_error, mean_absolute_error

import pandas as pd

import numpy as np

# Get the squared residuals from the mean model (AR(1)) as a proxy for realized
volatility

squared_residuals = residuals**2

# Get the conditional volatility from each volatility model and square it to get
conditional variance

arch_conditional_variance = arch_model_fit.conditional_volatility**2

garch_conditional_variance = garch_model_fit.conditional_volatility**2

earch_conditional_variance = earch_model_fit.conditional_volatility**2

```

```
# Align the squared residuals with the conditional variance series (they should have the same index)
```

```
# This step is for safety, as the conditional_variance should align with the data used to fit the model (residuals)
```

```
aligned_squared_residuals_arch =  
squared_residuals[arch_conditional_variance.index]
```

```
aligned_squared_residuals_garch =  
squared_residuals[garch_conditional_variance.index]
```

```
aligned_squared_residuals_earch =  
squared_residuals[earch_conditional_variance.index]
```

```
# Calculate MSE and MAE for each volatility model
```

```
mse_arch = mean_squared_error(aligned_squared_residuals_arch,  
arch_conditional_variance)
```

```
mae_arch = mean_absolute_error(aligned_squared_residuals_arch,  
arch_conditional_variance)
```

```
rmse_arch = np.sqrt(mse_arch) # Calculate RMSE for volatility
```

```
mse_garch = mean_squared_error(aligned_squared_residuals_garch,  
garch_conditional_variance)
```

```
mae_garch = mean_absolute_error(aligned_squared_residuals_garch,  
garch_conditional_variance)
```

```
rmse_garch = np.sqrt(mse_garch) # Calculate RMSE for volatility
```

```
mse_earch = mean_squared_error(aligned_squared_residuals_earch,  
earch_conditional_variance)
```

```
mae_earch = mean_absolute_error(aligned_squared_residuals_earch,  
earch_conditional_variance)
```

```
rmse_earch = np.sqrt(mse_earch) # Calculate RMSE for volatility
```

```
# Create a comparison table
```

```

volatility_error_data = {

    'Metric': ['MSE (Volatility)', 'MAE (Volatility)', 'RMSE (Volatility)'],

    'ARCH(1)': [mse_arch, mae_arch, rmse_arch],

    'GARCH(1,1)': [mse_garch, mae_garch, rmse_garch],

    'EARCH(1,1)': [mse_earch, mae_earch, rmse_earch]

}

volatility_error_df = pd.DataFrame(volatility_error_data)

# Display the comparison table

display(volatility_error_df)

# Note about other requested metrics for volatility analysis

print("\nNote on other requested metrics for volatility models:")

print("- AIC and BIC are model fit criteria, not typically used for comparing volatility
forecast accuracy against realized volatility.")

print("- MASE, MAPE, and Theil's U are generally used for evaluating point forecasts
of the time series level, not the volatility forecasts (which are measures of spread).")

import matplotlib.pyplot as plt

import pandas as pd

# Get the original silver prices

original_prices = silver_df['Silver']

# Get the fitted values from the AR(1) model

ar_fitted_values = model_fit.fittedvalues

# Get the conditional volatility from the EARCH(1,1) model

earch_conditional_volatility = earch_model_fit.conditional_volatility

# Create a figure with two subplots

fig, axes = plt.subplots(2, 1, figsize=(14, 10), sharex=True)

```

```

# Plot original prices and AR(1) fitted values in the top subplot

axes[0].plot(original_prices.index, original_prices, label='Original Silver Price')

axes[0].plot(ar_fitted_values.index, ar_fitted_values, label='AR(1) Fitted Values',
color='YELLOW', linestyle='--')

axes[0].set_ylabel("Price")

axes[0].set_title("Silver Price and AR(1) Fitted Values")

axes[0].legend()

axes[0].grid(True)

# Plot EARCH(1,1) Conditional Volatility in the bottom subplot

axes[1].plot(earch_conditional_volatility.index, earch_conditional_volatility,
label='EARCH(1,1) Conditional Volatility', color='green')

axes[1].set_ylabel("Volatility")

axes[1].set_title("EARCH(1,1) Conditional Volatility")

axes[1].legend()

axes[1].grid(True)

plt.xlabel("Date")

plt.tight_layout()

plt.show()

# Create a flowchart object

flowchart = Digraph(comment='Silver Price Analysis Methodology Flowchart',
format='png', engine='dot')

# Set graph attributes for a professional and academic look

flowchart.attr(size='8,10.5', rankdir='TB', dpi='300', nodesep='0.8', ranksep='0.8') #
Adjusted size for typical paper, increased spacing

flowchart.attr('node', shape='box', style='rounded', fontname='Arial', fontsize='10') #
Standard box shape, rounded corners, common academic font, adjusted size

```

```
flowchart.attr('edge', arrowhead='vee', arrowsize='0.8') # Standard arrowhead style,  
adjusted size
```

```
# Define nodes with a consistent box shape
```

```
flowchart.node("A", "Data Collection\n(Yahoo Finance)")
```

```
flowchart.node("B", "Preprocessing\n(Handle Missing Values, Log>Returns)")
```

```
flowchart.node("C", "Exploratory Analysis\n(Plots, ACF, PACF)")
```

```
flowchart.node("D", "Mean Modeling\n(AR, ARIMA)")
```

```
flowchart.node("E", "Residual Diagnostics\n(ARCH-LM Test)")
```

```
flowchart.node("F", "Volatility Modeling\n(ARCH, GARCH, EGARCH)")
```

```
flowchart.node("G", "Model Evaluation\n(Error Metrics, IC, Diagnostics)")
```

```
flowchart.node("H", "Model Selection\n(Best Fit for Mean & Volatility)")
```

```
# Define edges to show the flow
```

```
flowchart.edges(['AB', 'BC', 'CD', 'DE', 'EF', 'FG', 'GH'])
```

```
# Add conditional paths based on Residual Diagnostics
```

```
flowchart.edge("E", "F", label="Heteroskedasticity Present", fontname='Arial',  
fontsize='9')
```

```
# Assuming if no heteroskedasticity, we would stop at Mean Modeling or proceed  
without Volatility Modeling
```

```
# flowchart.edge("E", "G", label="No Heteroskedasticity", constraint='false',  
fontname='Arial', fontsize='9') # Example of an alternative path
```

```
# Render the flowchart to a file
```

```
flowchart.render('silver_price_analysis_flowchart_academic', view=True)
```

```
print("Flowchart generated as 'silver_price_analysis_flowchart_academic.png'")
```

FEEDBACK FORM

THIAGARAJAR SCHOOL OF MANAGEMENT
Pambanswamy Nagar, Thirupparankundram
Madurai-625 005

SIP FEEDBACK FORM

Name of the Student: Nandha Kumar T

Reg No: 2411051

Name of the SIP Organization: RK Global Shares and Securities

SIP Domain: Finance – Stock & Equity Market

Project Title: “Predicting Silver Price Trends Using AI and Time Series Models”

SIP Mentor’s Name: Mr. Suresh Ram

SIP Mentor’s Contact details: +91 91500 50040

Kindly tick the opinion that best describes your considered opinion:

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Parameters	1	2	3	4	5
1. His/Her Reporting to the work					✓
2. His/Her Sense of Responsibility					✓
3. His/Her Knowledge in the area			✓		
4. His/Her approach towards Project Objective				✓	
5. His/Her Research work				✓	
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7. His/her Contribution to the Organization					✓
8. His/Her Analytical Ability					✓
9. His/Her ability to finish work before the deadline					✓
10. His/ Her ability to correlate Theoretical concepts in practical experience				✓	
11. His/Her Effective Utilization of Time, resources for the Project				✓	
12. His/Her initiatives in the Organization				✓	
13. His/Her Leadership Skills					✓
14. His/Her Discipline in the Workspace				✓	
15. His/Her Overall Performance during the internship					✓

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Any Other Comments if specific also whether he/she is recommended for a Pre-Placement Offer: No



[Handwritten Signature]

Signature of SIP Mentor

ORIGINALITY REPORT

Nandha Kumar T

SIP PAPER DRAFT

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